

Find the greatest common factor of 16 and 40.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q5/5

801

QUESTION ID

S21-M-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q5/5

SET 21 • 6.NS.B.4

BACKWARD 1

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 18.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q1/5

803

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, 3, 6, 9, 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q1/5

True or false?

7 is a factor of 42.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q2/5

804

QUESTION ID

S21-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q2/5

Which number is NOT a factor of 36?

- A.** 4
- B.** 8
- C.** 9
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q3/5

805

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q3/5

True or false?

5 is a factor of 32.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q4/5

806

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q4/5

List all the factors of 24.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q5/5

807

QUESTION ID

S21-B1-Q5

TYPE

Short answer

CORRECT ANSWER

**1, 2, 3, 4, 6, 8, 12,
24**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 6.NS.B.4

BACKWARD 2

Prime and composite numbers • Grade 4 • Operations & Algebraic Thinking •
4.OA.B.4

True or false?

13 is a prime number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q1/5

809

QUESTION ID

S21-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q1/5

Which number is composite?

- A.** 2
- B.** 11
- C.** 17
- D.** 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q2/5

810

QUESTION ID

S21-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q2/5

True or false?

1 is a prime number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q3/5

811

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q3/5

Is 29 prime or composite?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q4/5

812

QUESTION ID

S21-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Prime

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q4/5

Which number is prime?

- A.** 27
- B.** 31
- C.** 33
- D.** 39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q5/5

813

QUESTION ID

S21-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 6.NS.B.4

BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$42 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q1/5

815

QUESTION ID

S21-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q1/5

Divide.

$$36 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q2/5

816

QUESTION ID

S21-B3-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q2/5

True or false?

$$45 \div 9 = 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q3/5

817

QUESTION ID

S21-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q3/5

$$56 \div 8 = ?$$

- A.** 6
- B.** 7
- C.** 8
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q4/5

818

QUESTION ID

S21-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$63 \div 9$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q5/5

819

QUESTION ID

S21-B3-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q5/5

819

SET 21 • 6.NS.B.4

FORWARD 1

Factor linear expressions using a common factor • Grade 7 • Expressions & Equations • 7.EE.A.1

Factor using the greatest common factor.

$$18x + 24$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q1/5

821

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6(3x + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q1/5

Factor using the greatest common factor.

$$20a - 35$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q2/5

822

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5(4a - 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q2/5

Which shows $16m + 40$ factored using the greatest common factor?

- A. $8(2m + 5)$
- B. $2(8m + 20)$
- C. $8(2m + 40)$
- D. $4(4m + 10)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q3/5

823

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $8(2m + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q3/5

Factor using the greatest common factor.

$$27x + 45y$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q4/5

824

QUESTION ID

S21-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$9(3x + 5y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q4/5

True or false?

$$14n - 21 = 7(2n - 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q5/5

825

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 6.NS.B.4

FORWARD 2

Solve equations using the distributive property • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$12x + 18 = 6(x + 5)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q1/5

827

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q1/5

Solve for x.

$$4(3x - 2) = 8(x + 1)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q2/5

828

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q2/5

Solve for x.

$$15x - 10 = 5(2x + 3)$$

- A. -1
- B. 1
- C. 5
- D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q3/5

829

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q3/5

Solve for x.

$$9(x + 2) = 6(x + 5)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q4/5

830

QUESTION ID

S21-F2-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q4/5

Solve for x.

$$10x + 25 = 5(x + 10)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q5/5

831

QUESTION ID

S21-F2-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q5/5

Find the least common multiple of two whole numbers

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Multiples	4.OA.B.4
BACKWARD 2	Multiplication facts	3.OA.C.7
FORWARD 1	Add and subtract rational numbers with unlike denominators	7.NS.A.1.d
FORWARD 2	Equations with fractions: clearing denominators	8.EE.C.7.b

SET 22 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 8.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q1/5

834

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q1/5

Find the least common multiple of 4 and 10.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q2/5

835

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q2/5

What is the least common multiple of 9 and 12?

- A.** 3
- B.** 36
- C.** 72
- D.** 108

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q3/5

836

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q3/5

Find the least common multiple of 5 and 7.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q4/5

837

QUESTION ID

S22-M-Q4

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q4/5

Find the least common multiple of 6 and 15.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q5/5

838

QUESTION ID

S22-M-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q5/5

SET 22 • 6.NS.B.4

BACKWARD 1

Multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five multiples of 6.

Multiples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q1/5

840

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6, 12, 18, 24, 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q1/5

True or false?

36 is a multiple of 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q2/5

841

QUESTION ID

S22-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q2/5

Which number is a multiple of both 4 and 6?

- A.** 8
- B.** 12
- C.** 18
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q3/5

842

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q3/5

True or false?

45 is a multiple of 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q4/5

843

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q4/5

What is the 4th multiple of 7?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q5/5

844

QUESTION ID

S22-B1-Q5

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 6.NS.B.4

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$6 \times 8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q1/5

846

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q1/5

Multiply.

$$9 \times 4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q2/5

847

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q2/5

True or false?

$$7 \times 5 = 35$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q3/5

848

QUESTION ID

S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q3/5

$$3 \times 12 = ?$$

- A.** 15
- B.** 33
- C.** 36
- D.** 39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q4/5

849

QUESTION ID

S22-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$5 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q5/5

850

QUESTION ID

S22-B2-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 6.NS.B.4

FORWARD 1

Add and subtract rational numbers with unlike denominators • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{5}{6} + \frac{3}{8}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q1/5

852

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−11/24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q1/5

Find the value.

$$\frac{3}{4} - \left(-\frac{1}{6}\right)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q2/5

853

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

11/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q2/5

$$-\frac{2}{3} - \frac{1}{4} = ?$$

A. $-\frac{11}{12}$

B. $-\frac{3}{7}$

C. $-\frac{1}{12}$

D. $\frac{5}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q3/5

854

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-\frac{11}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q3/5

Find the value.

$$-1\frac{1}{2} + \frac{5}{6}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q4/5

855

QUESTION ID

S22-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$-2/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q4/5

True or false?

$$\frac{3}{10} + \left(-\frac{2}{15}\right) = \frac{1}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q5/5

856

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q5/5

SET 22 • 6.NS.B.4

FORWARD 2

Equations with fractions: clearing denominators • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$\frac{x}{6} + \frac{x}{8} = 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q1/5

858

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q1/5

Solve for x.

$$\frac{x}{4} - 1 = \frac{x}{6} + 2$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q2/5

859

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q2/5

Solve for x.

$$\frac{2x}{3} + \frac{x}{2} = 14$$

- A. 7
- B. 12
- C. $\frac{84}{5}$
- D. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q3/5

860

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{x}{5} + 3 = \frac{x}{2}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q4/5

861

QUESTION ID

S22-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q4/5

Solve for x.

$$\frac{(x+1)}{4} = \frac{(x-2)}{3}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q5/5

862

QUESTION ID

S22-F2-Q5

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q5/5

Use the distributive property to write a sum as the GCF times a sum

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Common factors	4.OA.B.4
BACKWARD 2	Distributive property with whole numbers	3.OA.B.5
BACKWARD 3	Parentheses and order of operations	5.OA.A.1
FORWARD 1	Factor and expand linear expressions	7.EE.A.1
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 23 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $45 + 20$ as a product of the GCF and a sum of two whole numbers.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q1/5

865

QUESTION ID

S23-M-Q1

TYPE

Short answer

CORRECT ANSWER

5(9 + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q1/5

Which expression is equal to $42 + 56$ and uses the greatest common factor?

- A. $2(21 + 28)$
- B. $14(3 + 4)$
- C. $14(3 + 8)$
- D. $7(6 + 8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q2/5

866

QUESTION ID

S23-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $14(3 + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q2/5

Use the greatest common factor to write $27 + 45$ as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q3/5

867

QUESTION ID

S23-M-Q3

TYPE

Short answer

CORRECT ANSWER

9(3 + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q3/5

True or false?

$$24 + 60 = 12(2 + 5)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q4/5

868

QUESTION ID

S23-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q4/5

Complete the equation.

$$30 + 75 = 15(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q5/5

869

QUESTION ID

S23-M-Q5

TYPE

Short answer

CORRECT ANSWER

**First blank: 2;
Second blank: 5**

GRADING NOTE 5 and 2 in either order is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q5/5

SET 23 • 6.NS.B.4

BACKWARD 1

Common factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

5 is a factor of both 45 and 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q1/5

871

QUESTION ID

S23-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q1/5

List the factors of 12 and the factors of 18.

Which factors do they share?

Shared factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q2/5

872

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1, 2, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q2/5

Which number is a factor of both 27 and 45?

- A.** 5
- B.** 6
- C.** 9
- D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q3/5

873

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q3/5

List all the factors of 42.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q4/5

874

QUESTION ID

S23-B1-Q4

TYPE

Short answer

CORRECT ANSWER

**1, 2, 3, 6, 7, 14, 21,
42**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q4/5

True or false?

6 is a factor of 56.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q5/5

875

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 6.NS.B.4

BACKWARD 2

Distributive property with whole numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$9 \times 6 = 9 \times 4 + 9 \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

877

QUESTION ID

S23-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

True or false?

$$4 \times (9 + 2) = 4 \times 9 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

878

QUESTION ID

S23-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

Which is equal to 6×12 ?

- A. $6 \times 10 + 2$
- B. $6 \times 10 \times 2$
- C. $6 \times 10 + 6 \times 2$
- D. $6 + 10 \times 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

879

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $6 \times 10 + 6 \times 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

$$5 \times 13 = 5 \times 10 + 5 \times \underline{\quad}$$

Blank: _____

ID S23-B2-Q4

880

S23-B2-Q4

Short answer

3

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

$$3 \times (4 + 5) = 3 \times 4 + 3 \times \underline{\quad}$$

Blank: _____

ID S23-B2-Q5

881

S23-B2-Q5

Short answer

5

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 6.NS.B.4

BACKWARD 3

Parentheses and order of operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$5 \times (9 + 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q1/5

883

QUESTION ID

S23-B3-Q1

TYPE

Short answer

CORRECT ANSWER

65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q1/5

Evaluate.

$$14 \times (3 + 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q2/5

884

QUESTION ID

S23-B3-Q2

TYPE

Short answer

CORRECT ANSWER

98

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q2/5

True or false?

$$9 \times (3 + 5) = 32$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q3/5

885

QUESTION ID

S23-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q3/5

$$12 \times (2 + 5) = ?$$

- A.** 17
- B.** 29
- C.** 60
- D.** 84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q4/5

886

QUESTION ID

S23-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q4/5

Evaluate.

$$(20 - 5) \times 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q5/5

887

QUESTION ID

S23-B3-Q5

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • 6.NS.B.4

FORWARD 1

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor.

$$12x - 18$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q1/5

889

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$6(2x - 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Expand.

$$-3(2x - 5)$$

Expanded form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q2/5

890

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$$-6x + 15$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q2/5

Which expression is equivalent to $8y + 20$?

- A. $4(2y + 20)$
- B. $8(y + 20)$
- C. $2(4y + 20)$
- D. $4(2y + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q3/5

891

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $4(2y + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q3/5

Factor.

$$15a + 10b - 5$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q4/5

892

QUESTION ID

S23-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$5(3a + 2b - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q4/5

True or false?

$$\frac{1}{2}(6x - 4) = 3x - 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q5/5

893

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 6.NS.B.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

How many solutions does the equation have?

$$4(x + 3) = 4x + 12$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q1/5

895

QUESTION ID

S23-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q1/5

How many solutions does the equation have?

$$2(3x - 1) = 6x + 5$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q2/5

896

QUESTION ID

S23-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q2/5

How many solutions does the equation have?

$$3(x + 2) = 2x + 9$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q3/5

897

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q3/5

What number goes in the box so that the equation has infinitely many solutions?

$$5(x - 2) = 5x - \square$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q4/5

898

QUESTION ID

S23-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q4/5

Which equation has no solution?

- A. $2(x + 4) = 2x + 5$
- B. $2(x + 4) = x + 8$
- C. $2(x + 4) = 2x + 8$
- D. $2(x + 4) = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q5/5

899

QUESTION ID

S23-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $2(x + 4) = 2x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q5/5

Use positive and negative numbers to represent quantities in context

MAIN	Grade 6 main question	6.NS.C.5
BACKWARD 1	Whole numbers as distances from 0	2.MD.B.6
BACKWARD 2	Increase and decrease situations	2.OA.A.1
FORWARD 1	Adding integers as moving on a number line	7.NS.A.1.b
FORWARD 2	The sign of the rate of change in context	8.F.B.4

SET 24 • 6.NS.C.5

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.5

A diver is 30 feet below sea level.

Write an integer to represent the diver's position.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q1/5

902

QUESTION ID

S24-M-Q1

TYPE

Short answer

CORRECT ANSWER

−30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q1/5

A company lost \$2,500 this month.

Write an integer to represent this change.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q2/5

903

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

−2,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q2/5

A bank account shows a withdrawal of \$45.

Which number represents this change?

- A. -45
- B. -4.5
- C. 0
- D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q3/5

904

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q3/5

The temperature rose 12 degrees.

Write an integer to represent this change.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q4/5

905

QUESTION ID

S24-M-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE +12 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q4/5

A hawk flies 120 feet above sea level. A submarine is 120 feet below sea level.

Write an integer for each position.

Hawk: _____

Submarine: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q5/5

906

QUESTION ID

S24-M-Q5

TYPE

Short answer

CORRECT ANSWER

**Hawk: 120;
Submarine: -120**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q5/5

SET 24 • 6.NS.C.5

BACKWARD 1

Whole numbers as distances from 0 • Grade 2 • Measurement & Data •
2.MD.B.6

What number does point A represent?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q1/5

908

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q1/5

How many units from 0 is the number 6?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q2/5

909

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q2/5

True or false?

On a number line, 9 is farther from 0 than 4 is.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q3/5

910

QUESTION ID

S24-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q3/5

Which point is 3 units from 0?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q4/5

911

QUESTION ID

S24-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q4/5

What number does point B represent?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q5/5

912

QUESTION ID

S24-B1-Q5

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 6.NS.C.5

BACKWARD 2

Increase and decrease situations • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

Kim had \$20. She spent \$8.

How much money does she have now?

Money: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q1/5

914

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

\$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q1/5

A plant was 12 cm tall. It grew 5 cm.

How tall is the plant now?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q2/5

915

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

17 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q2/5

Which situation describes a decrease?

- A. A plant grows 4 inches.
- B. Tom earns \$10.
- C. A team gains 5 yards.
- D. A balloon loses 6 feet of height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q3/5

916

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A balloon loses 6 feet of height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q3/5

True or false?

"The temperature fell 7 degrees" describes a decrease.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q4/5

917

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q4/5

Which word describes an increase in money?

- A.** Deposit
- B.** Loss
- C.** Drop
- D.** Withdrawal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q5/5

918

QUESTION ID

S24-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Deposit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 6.NS.C.5

FORWARD 1

Adding integers as moving on a number line • Grade 7 • The Number System •
7.NS.A.1.b

The temperature is -4°F . It rises 9°F .

What is the new temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q1/5

920

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q1/5

A submarine is at -120 feet. It rises 45 feet.

What is its new position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q2/5

921

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-75 feet

GRADING NOTE 75 feet below sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q2/5

On a number line, where is $6 + (-4)$ compared to 6?

- A. 4 units to the right of 6
- B. 4 units to the left of 6
- C. At 0
- D. 6 units to the left of 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q3/5

922

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4 units to the left of 6

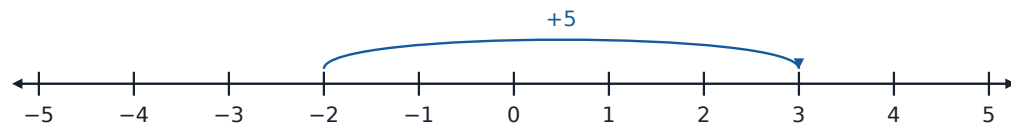
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q3/5

The number line shows $-2 + 5$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q4/5

923

QUESTION ID

S24-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q4/5

True or false?

A \$30 debt plus a \$30 deposit makes a balance of \$0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q5/5

924

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q5/5

SET 24 • 6.NS.C.5

FORWARD 2

The sign of the rate of change in context • Grade 8 • Functions • 8.F.B.4

A tank holds 60 gallons. It drains 3 gallons per minute.

Write a function for the amount A after t minutes.

$A =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q1/5

926

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$A = 60 - 3t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q1/5

An airplane at 12,000 feet descends 800 feet per minute.

What is the rate of change of its altitude?

- A.** $-12,000$ feet per minute **B.** -800 feet per minute
C. 800 feet per minute **D.** $12,000$ feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q2/5

927

QUESTION ID

S24-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. -800 feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q2/5

The table shows a linear function.

What is the rate of change in degrees per hour?

Hours	0	2	4
Temperature (°F)	5	1	-3

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q3/5

928

QUESTION ID

S24-F2-Q3

TYPE

Short answer

CORRECT ANSWER

-2 degrees per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q3/5

A diver starts at -10 feet and descends 4 feet per second.

Write a function for the diver's position d after t seconds.

$d =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q4/5

929

QUESTION ID

S24-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$d = -10 - 4t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q4/5

True or false?

A line through $(0, 8)$ and $(4, 0)$ has a rate of change of -2 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q5/5

930

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q5/5

Explain the meaning of 0 in a real-world situation

MAIN	Grade 6 main question	6.NS.C.5
BACKWARD 1	Zero as the starting point on a number line	2.MD.B.6
BACKWARD 2	A change of zero means no change	1.OA.D.8
FORWARD 1	Opposite quantities combine to make 0	7.NS.A.1.a
FORWARD 2	Interpret the initial value (the value when $x = 0$)	8.F.B.4

SET 25 • 6.NS.C.5

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.5

On a Celsius thermometer, what does 0°C represent?

- A. No temperature at all
- B. The temperature at which water freezes
- C. The hottest possible temperature
- D. The temperature at which water boils

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q1/5

933

QUESTION ID

S25-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

**B. The temperature
at which water
freezes**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q1/5

When numbers describe elevation, what does 0 represent?

- A.** Sea level
- B.** No elevation at all
- C.** The top of a mountain
- D.** The ocean floor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q2/5

934

QUESTION ID

S25-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q2/5

A bank account has a balance of \$0.

What does this mean?

- A. The account has no money and owes nothing.
- B. The account owes money.
- C. The account has money in it.
- D. The account earns interest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q3/5

935

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

**A. The account has
no money and owes
nothing.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q3/5

A football team's change in yards on a play is 0.

What does 0 mean?

- A. The team scored.
- B. The team gained 10 yards.
- C. The team neither gained nor lost yards.
- D. The team lost all its yards.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q4/5

936

QUESTION ID

S25-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The team neither gained nor lost yards.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q4/5

A scale shows that your weight changed by 0 pounds since last month.

What does 0 mean in this situation?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q5/5

937

QUESTION ID

S25-M-Q5

TYPE

Short answer

CORRECT ANSWER

The weight did not change: there was no gain and no loss.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q5/5

SET 25 • 6.NS.C.5

BACKWARD 1

Zero as the starting point on a number line • Grade 2 • Measurement & Data •
2.MD.B.6

True or false?

On a number line, 0 is the starting point for measuring distance.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q1/5

939

QUESTION ID

S25-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q1/5

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q2/5

940

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q2/5

How far is 0 from itself on a number line?

- A.** 0 units
- B.** 1 unit
- C.** 2 units
- D.** 10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q3/5

941

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q3/5

How far is 5 from 0 on a number line?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q4/5

942

QUESTION ID

S25-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q4/5

True or false?

A ruler starts measuring at 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q5/5

943

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 6.NS.C.5

BACKWARD 2

A change of zero means no change • Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

What number goes in the box?

$$8 + \square = 8$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q1/5

945

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q1/5

What number goes in the box?

$$15 - \square = 15$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q2/5

946

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q2/5

True or false?

A plant was 12 cm tall last week and is 12 cm tall now. It grew 0 cm.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q3/5

947

QUESTION ID

S25-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q3/5

A score went from 45 to 45.

What was the change?

- A.** 0
- B.** 1
- C.** 45
- D.** 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q4/5

948

QUESTION ID

S25-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q4/5

What number goes in the box?

$$\square + 23 = 23$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q5/5

949

QUESTION ID

S25-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 6.NS.C.5

FORWARD 1

Opposite quantities combine to make 0 • Grade 7 • The Number System •
7.NS.A.1.a

Maya earns \$25 and then spends \$25.

What is the total change in her money?

Total change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q1/5

951

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

\$0

GRADING NOTE No change

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q1/5

True or false?

$$-8 + 8 = 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q2/5

952

QUESTION ID

S25-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q2/5

Which sum is equal to 0?

- A. $-3.5 + 3.5$
- B. $-3.5 + 0$
- C. $-3.5 + (-3.5)$
- D. $3.5 + 3.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q3/5

953

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-3.5 + 3.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q3/5

A balloon rises 40 feet and then falls 40 feet.

What is its total change in height?

Total change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q4/5

954

QUESTION ID

S25-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0 feet

GRADING NOTE No change

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q4/5

What number added to 14.2 gives 0?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q5/5

955

QUESTION ID

S25-F1-Q5

TYPE

Short answer

CORRECT ANSWER

−14.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q5/5

SET 25 • 6.NS.C.5

FORWARD 2

Interpret the initial value (the value when $x = 0$) • Grade 8 • Functions •
8.F.B.4

The water in a tank is $W = 60 - 3t$, where t is minutes.

What does 60 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q1/5

957

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**The starting amount:
the tank holds 60
gallons at $t = 0$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q1/5

A hiker's elevation is $E = -40 + 15t$, where t is hours.

What does -40 represent?

- A. The hiker starts 40 m below sea level.
- B. The hiker ends 40 m below sea level.
- C. The hiker climbs 40 m each hour.
- D. The hike lasts 40 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q2/5

958

QUESTION ID

S25-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

**A. The hiker starts
40 m below sea
level.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q2/5

A phone's battery charge is $B = 100 - 8h$ after h hours.

What is B when $h = 0$, and what does it mean?

Value: _____

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q3/5

959

QUESTION ID

S25-F2-Q3

TYPE

Short answer

CORRECT ANSWER

**Value: 100; Meaning:
the battery starts
fully charged (100%)
at 0 hours**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q3/5

A savings account has $y = 25x + 150$ dollars after x weeks.

What does 150 represent?

- A.** The starting amount
- B.** The amount saved each week
- C.** The number of weeks
- D.** The amount after 25 weeks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q4/5

960

QUESTION ID

S25-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The starting amount

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q4/5

True or false?

For the function $y = 4x$, the output is 0 when the input is 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q5/5

961

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q5/5

Find opposites; the opposite of the opposite of a number

MAIN

Grade 6 main question

6.NS.C.6.a

BACKWARD 1

Distance from 0 on a number line

2.MD.B.6

BACKWARD 2

Positive and negative numbers in context

6.NS.C.5

FORWARD 1

Subtract by adding the opposite

7.NS.A.1.c

FORWARD 2

Solve equations by adding opposites to both sides

8.EE.C.7.b

SET 26 • 6.NS.C.6.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of -7 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q1/5

964

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q1/5

Simplify.

$-(-12)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q2/5

965

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

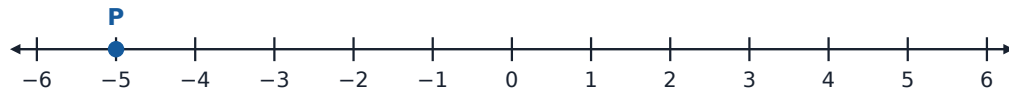
12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q2/5

Which number is the opposite of the number at point P?



A. -5

B. 0

C. $\frac{1}{5}$

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q3/5

966

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q3/5

True or false?

The opposite of 0 is 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q4/5

967

QUESTION ID

S26-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q4/5

What is the opposite of the opposite of 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q5/5

968

QUESTION ID

S26-M-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q5/5

SET 26 • 6.NS.C.6.a

BACKWARD 1

Distance from 0 on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How far is point A from 0?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q1/5

970

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q1/5

How many units from 0 is the number 8?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q2/5

971

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q2/5

Which number is 2 units from 0 on this number line?



A. 0

B. 2

C. 4

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q3/5

972

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q3/5

What number does point K represent?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q4/5

973

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q4/5

True or false?

The distance from 0 to 7 is 7 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q5/5

974

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 6.NS.C.6.a

BACKWARD 2

Positive and negative numbers in context • Grade 6 • The Number System •
6.NS.C.5

True or false?

–5 can represent 5 degrees below zero.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q1/5

976

QUESTION ID

S26-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q1/5

Write an integer for a gain of 12 yards.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q2/5

977

QUESTION ID

S26-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE +12 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q2/5

Which integer represents 20 feet below sea level?

- A.** -20
- B.** -2
- C.** 0
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q3/5

978

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q3/5

Write an integer for a loss of \$7.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q4/5

979

QUESTION ID

S26-B2-Q4

TYPE

Short answer

CORRECT ANSWER

-7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q4/5

True or false?

A deposit of \$15 is written as -15 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q5/5

980

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 6.NS.C.6.a

FORWARD 1

Subtract by adding the opposite • Grade 7 • The Number System • 7.NS.A.1.c

Rewrite $5 - 8$ as an addition expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q1/5

982

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$5 + (-8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q1/5

Find the value.

$$-3 - (-7)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q2/5

983

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q2/5

True or false?

$$6 - (-2) = 6 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q3/5

984

QUESTION ID

S26-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q3/5

Which expression is equal to $-4 - 9$?

- A. $4 + 9$
- B. $4 - (-9)$
- C. $-4 + 9$
- D. $-4 + (-9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q4/5

985

QUESTION ID

S26-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $-4 + (-9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q4/5

Find the value.

$$2.5 - 6$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q5/5

986

QUESTION ID

S26-F1-Q5

TYPE

Short answer

CORRECT ANSWER

-3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q5/5

SET 26 • 6.NS.C.6.a

FORWARD 2

Solve equations by adding opposites to both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$5x - 7 = 2x + 8$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q1/5

988

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q1/5

Solve for x.

$$-3x + 4 = 3x - 8$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q2/5

989

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q2/5

Solve for x.

$$6 - 2x = 4x - 12$$

- A.** -3
- B.** 1
- C.** 3
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q3/5

990

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q3/5

Solve for x.

$$7x + 3 = -2x + 21$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q4/5

991

QUESTION ID

S26-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q4/5

Solve for x.

$$4(x - 1) = 2x + 10$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q5/5

992

QUESTION ID

S26-F2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q5/5

Use signs of coordinates to identify quadrants

MAIN	Grade 6 main question	6.NS.C.6.b
BACKWARD 1	Axes, origin, and ordered pairs	5.G.A.1
BACKWARD 2	Negative numbers are on the opposite side of 0	6.NS.C.6.a
BACKWARD 3	Reading points in the first quadrant	5.G.A.2
FORWARD 1	Signs of products of rational numbers	7.NS.A.2.a
FORWARD 2	Transformations and the quadrant of the image	8.G.A.3

SET 27 • 6.NS.C.6.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.b

In which quadrant is the point $(-4, 7)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q1/5

995

QUESTION ID

S27-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Quadrant II

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q1/5

In which quadrant is the point $(3, -8)$?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q2/5

996

QUESTION ID

S27-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q2/5

In which quadrant is the point $(-2.5, -6)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q3/5

997

QUESTION ID

S27-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Quadrant III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q3/5

A point has a negative x-coordinate and a positive y-coordinate.

In which quadrant is it?

Quadrant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q4/5

998

QUESTION ID

S27-M-Q4

TYPE

Short answer

CORRECT ANSWER

Quadrant II

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q4/5

Which point is in Quadrant III?

- A.** $(-5, -1)$
- B.** $(5, 1)$
- C.** $(5, -1)$
- D.** $(-5, 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q5/5

999

QUESTION ID

S27-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $(-5, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q5/5

SET 27 • 6.NS.C.6.b

BACKWARD 1

Axes, origin, and ordered pairs • Grade 5 • Geometry • 5.G.A.1

True or false?

The origin is the point (0, 0).

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q1/5

1001

QUESTION ID

S27-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q1/5

In the ordered pair (5, 2), which number tells how far to move along the x-axis?

- A. 2
- B. 3
- C. 5
- D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q2/5

1002

QUESTION ID

S27-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q2/5

What is the name of the point where the x-axis and the y-axis meet?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q3/5

1003

QUESTION ID

S27-B1-Q3

TYPE

Short answer

CORRECT ANSWER

The origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q3/5

True or false?

In the ordered pair (3, 8), the y-coordinate is 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q4/5

1004

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q4/5

Which axis is horizontal?

- A.** The x-axis
- B.** The y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q5/5

1005

QUESTION ID

S27-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. The x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 6.NS.C.6.b

BACKWARD 2

Negative numbers are on the opposite side of 0 • Grade 6 • The Number System • 6.NS.C.6.a

True or false?

On a horizontal number line, -4 is to the left of 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q1/5

1007

QUESTION ID

S27-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q1/5

On a vertical number line, where is -3 ?

- A.** Above 0
- B.** Above 3
- C.** Below 0
- D.** At 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q2/5

1008

QUESTION ID

S27-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Below 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q2/5

True or false?

On a vertical number line, 5 is below 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q3/5

1009

QUESTION ID

S27-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q3/5

On a horizontal number line, is 2.5 to the left or the right of 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q4/5

1010

QUESTION ID

S27-B2-Q4

TYPE

Short answer

CORRECT ANSWER

To the right of 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q4/5

Which number is to the left of 0 on a horizontal number line?

- A. -6
- B. 0
- C. $\frac{1}{2}$
- D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q5/5

1011

QUESTION ID

S27-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. -6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q5/5

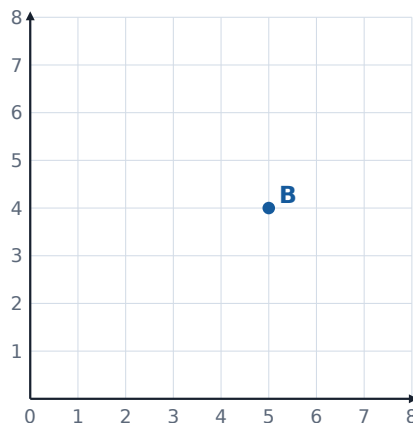
SET 27 • 6.NS.C.6.b

BACKWARD 3

Reading points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q1/5

1013

QUESTION ID

S27-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

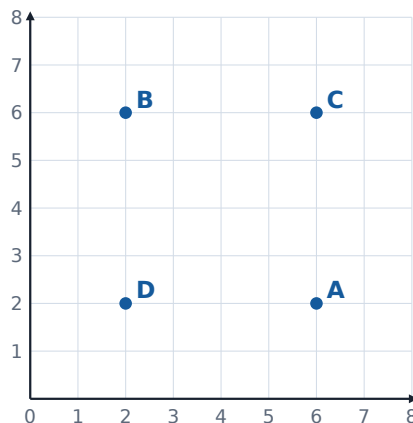
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q1/5

Which point is located at (2, 6)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q2/5

1014

QUESTION ID

S27-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

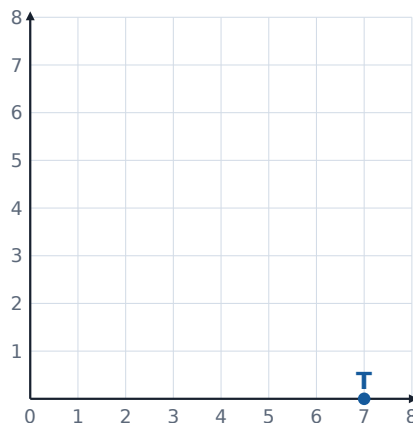
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q2/5

What are the coordinates of point T?

T = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q3/5

1015

QUESTION ID

S27-B3-Q3

TYPE

Short answer

CORRECT ANSWER

(7, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

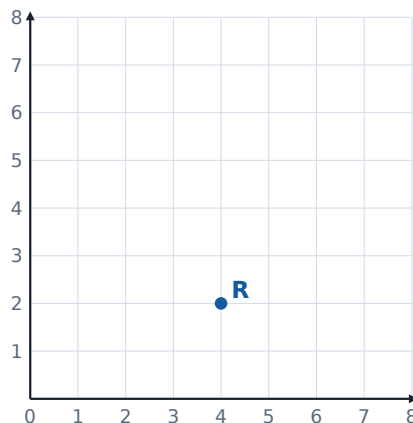
ID S27-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q3/5

True or false?

Point R is located at (2, 4).

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q4/5

1016

QUESTION ID

S27-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

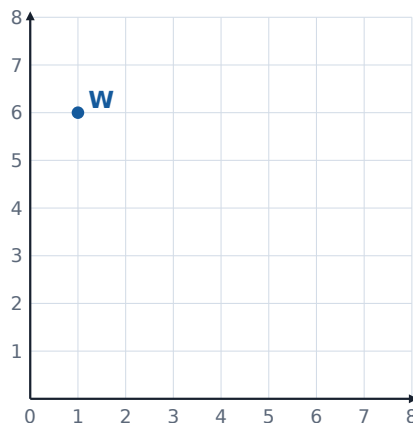
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q4/5

What are the coordinates of point W?

W = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q5/5

1017

QUESTION ID

S27-B3-Q5

TYPE

Short answer

CORRECT ANSWER

(1, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • 6.NS.C.6.b

FORWARD 1

Signs of products of rational numbers • Grade 7 • The Number System •
7.NS.A.2.a

The point (x, y) is in Quadrant II.
Is the product xy positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q1/5

1019

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q1/5

The point (x, y) is in Quadrant III.

Is the product xy positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q2/5

1020

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q2/5

For the point (x, y) , the product xy is positive and x is negative.

In which quadrant is the point?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q3/5

1021

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Quadrant III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q3/5

True or false?

If (x, y) is in Quadrant IV, then the product xy is negative.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q4/5

1022

QUESTION ID

S27-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q4/5

Find the product of the coordinates of the point $(-3, 5)$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q5/5

1023

QUESTION ID

S27-F1-Q5

TYPE

Short answer

CORRECT ANSWER

-15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q5/5

SET 27 • 6.NS.C.6.b

FORWARD 2

Transformations and the quadrant of the image • Grade 8 • Geometry •
8.G.A.3

The point (2, 5) is rotated 180° about the origin.

What are the coordinates of the image, and in which quadrant is it?

Image: _____

Quadrant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q1/5

1025

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**Image: $(-2, -5)$;
Quadrant: III**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q1/5

A point in Quadrant II is reflected across the y-axis.

In which quadrant is the image?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q2/5

1026

QUESTION ID

S27-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Quadrant I

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q2/5

The point $(-3, -4)$ is rotated 90° counterclockwise about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q3/5

1027

QUESTION ID

S27-F2-Q3

TYPE

Short answer

CORRECT ANSWER

(4, -3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q3/5

True or false?

Translating (1, 2) 5 units to the left moves it into Quadrant II.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q4/5

1028

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q4/5

The point $(4, -1)$ is reflected across the x-axis.

In which quadrant is the image?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q5/5

1029

QUESTION ID

S27-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Quadrant I

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q5/5

Reflect points across one or both axes

MAIN	Grade 6 main question	6.NS.C.6.b
BACKWARD 1	Opposite numbers	6.NS.C.6.a
BACKWARD 2	Lines of symmetry	4.G.A.3
BACKWARD 3	Reading coordinates in the first quadrant	5.G.A.2
FORWARD 1	Distance between a point and its reflection	7.NS.A.1.c
FORWARD 2	Reflections and rotations with coordinates	8.G.A.3

SET 28 • 6.NS.C.6.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.b

The point $(3, -5)$ is reflected across the x-axis.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q1/5

1032

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q1/5

The point $(-2, 6)$ is reflected across the y -axis.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q2/5

1033

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

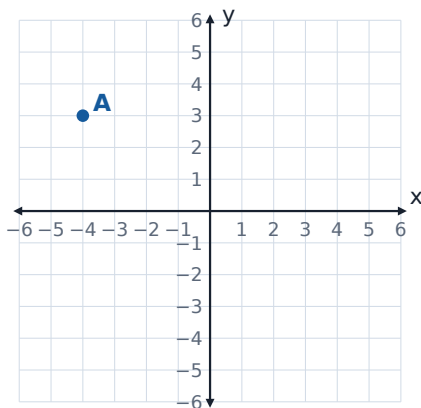
ID S28-M-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q2/5

Point A is reflected across the y-axis.

What are the coordinates of its image?

Image: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q3/5

1034

QUESTION ID

S28-M-Q3

TYPE

Short answer

CORRECT ANSWER

(4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q3/5

Which point is the reflection of $(-7, -2)$ across the x-axis?

- A. $(7, 2)$
- B. $(-2, -7)$
- C. $(-7, 2)$
- D. $(7, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q4/5

1035

QUESTION ID

S28-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(-7, 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q4/5

Point B(5, 1) is reflected across the y-axis and then across the x-axis.

What are the coordinates of the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q5/5

1036

QUESTION ID

S28-M-Q5

TYPE

Short answer

CORRECT ANSWER

(-5, -1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q5/5

SET 28 • 6.NS.C.6.b

BACKWARD 1

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 6?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q1/5

1038

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q1/5

True or false?

The opposite of -9 is 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q2/5

1039

QUESTION ID

S28-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q2/5

Which pair of numbers are opposites?

- A. 0 and 4
- B. -4 and -4
- C. 4 and $\frac{1}{4}$
- D. -4 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q3/5

1040

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. -4 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q3/5

What is the opposite of -2.5 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q4/5

1041

QUESTION ID

S28-B1-Q4

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q4/5

True or false?

$$-(-3) = -3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q5/5

1042

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 6.NS.C.6.b

BACKWARD 2

Lines of symmetry • Grade 4 • Geometry • 4.G.A.3

True or false?

A line of symmetry divides a figure into two matching halves.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q1/5

1044

QUESTION ID

S28-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q1/5

How many lines of symmetry does a square have?

- A.** 0
- B.** 1
- C.** 2
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q2/5

1045

QUESTION ID

S28-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 4

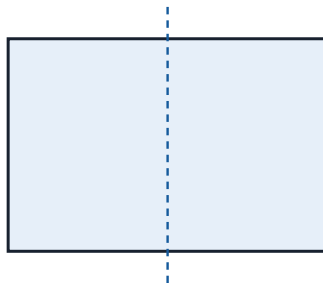
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q2/5

True or false?

The dashed line is a line of symmetry of the rectangle.



- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q3/5

1046

QUESTION ID

S28-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q3/5

Which letter has a vertical line of symmetry?

- A.** F
- B.** A
- C.** J
- D.** R

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q4/5

1047

QUESTION ID

S28-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q4/5

How many lines of symmetry does a rectangle that is not a square have?

Lines: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q5/5

1048

QUESTION ID

S28-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q5/5

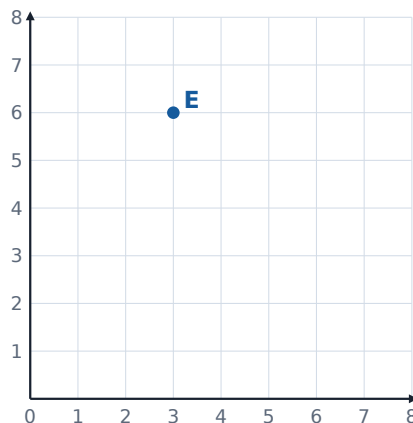
SET 28 • 6.NS.C.6.b

BACKWARD 3

Reading coordinates in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point E?

E = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q1/5

1050

QUESTION ID

S28-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 6)

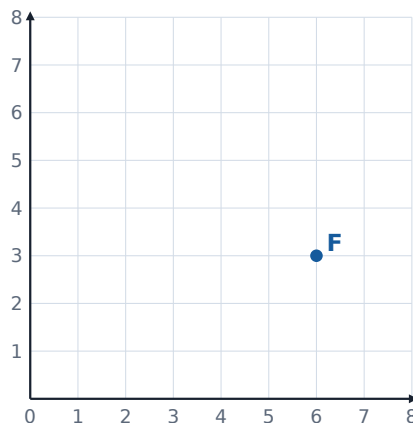
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q1/5

What are the coordinates of point F?

F = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q2/5

1051

QUESTION ID

S28-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(6, 3)

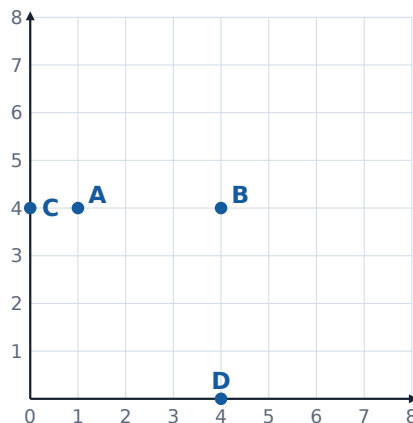
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q2/5

Which point is located at (4, 0)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q3/5

1052

QUESTION ID

S28-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

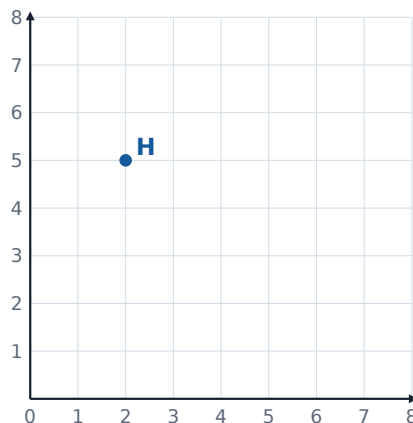
ID S28-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q3/5

True or false?

Point H is located at (5, 2).

- A. True
- B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q4/5

1053

QUESTION ID

S28-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

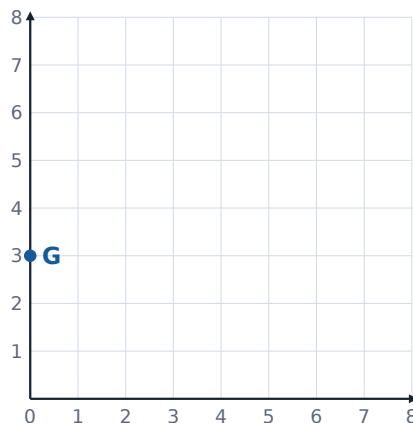
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q4/5

What are the coordinates of point G?

G = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q5/5

1054

QUESTION ID

S28-B3-Q5

TYPE

Short answer

CORRECT ANSWER

(0, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q5/5

SET 28 • 6.NS.C.6.b

FORWARD 1

Distance between a point and its reflection • Grade 7 • The Number System •
7.NS.A.1.c

The point $(3, -4.5)$ is reflected across the x-axis.

How far apart are the point and its image?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q1/5

1056

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q1/5

The point $(-6.5, 2)$ is reflected across the y -axis.

How far apart are the point and its image?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q2/5

1057

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q2/5

The point $(1, -2\frac{1}{2})$ is reflected across the x-axis.

How far apart are the point and its image?

- A. -5
- B. 0
- C. $2\frac{1}{2}$
- D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q3/5

1058

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q3/5

A point in Quadrant I and its reflection across the y-axis are 11 units apart. The point's y-coordinate is 7.

What is the point's x-coordinate?

X = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q4/5

1059

QUESTION ID

S28-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q4/5

True or false?

The point $(-4, 3)$ and its reflection across the x-axis are 6 units apart.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q5/5

1060

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q5/5

SET 28 • 6.NS.C.6.b

FORWARD 2

Reflections and rotations with coordinates • Grade 8 • Geometry • 8.G.A.3

A triangle has a vertex at $(2, -3)$. The triangle is reflected across the y -axis.

What are the coordinates of the image of that vertex?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q1/5

1062

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(-2, -3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q1/5

Which transformation is described by $(x, y) \rightarrow (x, -y)$?

- A. A rotation of 180°
- B. A reflection across the x-axis
- C. A translation
- D. A reflection across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q2/5

1063

QUESTION ID

S28-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

**B. A reflection
across the x-axis**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q2/5

The point $(4, 1)$ is rotated 180° about the origin.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q3/5

1064

QUESTION ID

S28-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$(-4, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

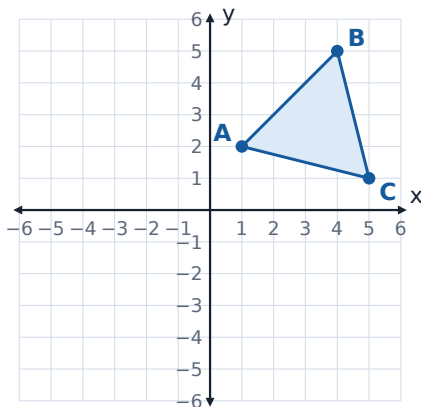
ID S28-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q3/5

Triangle ABC is reflected
across the x-axis.

What are the coordinates of
B'?

B' = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q4/5

1065

QUESTION ID

S28-F2-Q4

TYPE

Short answer

CORRECT ANSWER

(4, -5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q4/5

True or false?

Reflecting $(-3, 5)$ across the y -axis gives $(3, 5)$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q5/5

1066

QUESTION ID

S28-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q5/5

Name the rational number at a point on a number line

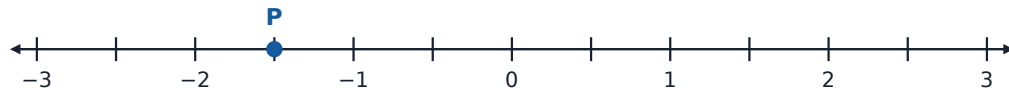
MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Fractions on a number line	3.NF.A.2
BACKWARD 2	Decimals on a number line	4.NF.C.6
BACKWARD 3	Opposites on a number line	6.NS.C.6.a
FORWARD 1	Addition shown on a number line	7.NS.A.1.b
FORWARD 2	Approximate irrational numbers on a number line	8.NS.A.2

SET 29 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q1/5

1069

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

-1.5

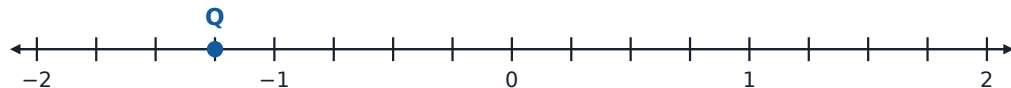
GRADING NOTE -1 1/2 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q1/5

Write the number at point Q as a mixed number.



Q = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q2/5

1070

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

$-1 \frac{1}{4}$

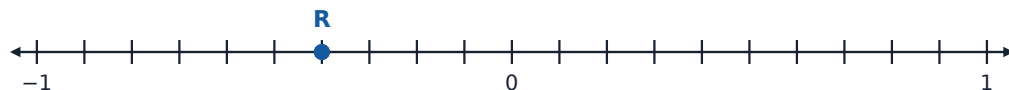
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q2/5

True or false?

Point R is located at -0.4 .



- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q3/5

1071

QUESTION ID

S29-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

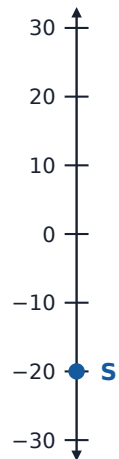
ID S29-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q3/5

The number line shows elevations in meters.

What elevation does point S show?

S = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q4/5

1072

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

–20 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q4/5

Write the number at point T as a fraction.



T = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q5/5

1073

QUESTION ID

S29-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

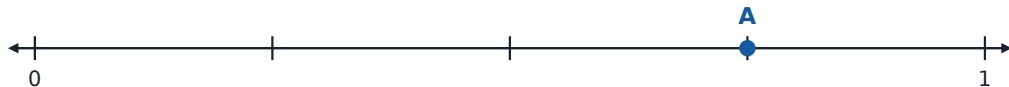
Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q5/5

SET 29 • 6.NS.C.6.c

BACKWARD 1

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction does point A represent?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q1/5

1075

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q1/5

What fraction does point B represent?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q2/5

1076

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

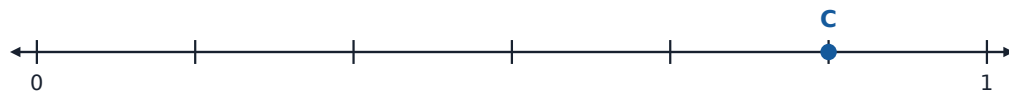
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q2/5

True or false?

Point C is located at $\frac{5}{6}$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q3/5

1077

QUESTION ID

S29-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q3/5

The space from 0 to 1 is split into equal parts.

How many equal parts are there?



A. 7

B. 8

C. 9

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q4/5

1078

QUESTION ID

S29-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

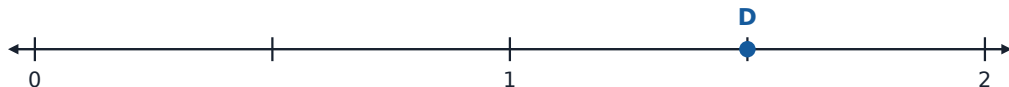
B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q4/5

Write the number at point D as a fraction.



D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q5/5

1079

QUESTION ID

S29-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3/2

GRADING NOTE 1 1/2 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

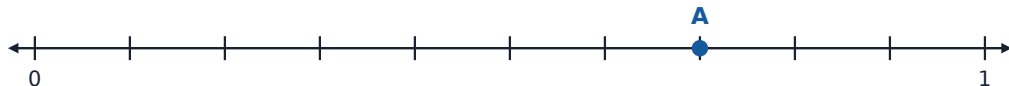
Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 6.NS.C.6.c

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal does point A represent?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q1/5

1081

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q1/5

What decimal does point B represent?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q2/5

1082

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2.4

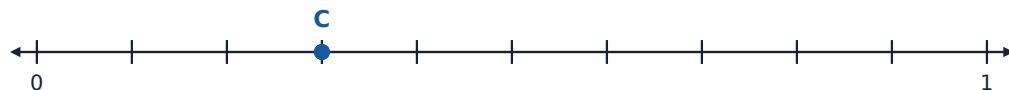
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q2/5

True or false?

Point C is located at 0.3.



- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q3/5

1083

QUESTION ID

S29-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q3/5

What decimal does point D represent?



A. 0.206

B. 0.26

C. 0.62

D. 2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q4/5

1084

QUESTION ID

S29-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q4/5

What decimal does point E represent?



E = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q5/5

1085

QUESTION ID

S29-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q5/5

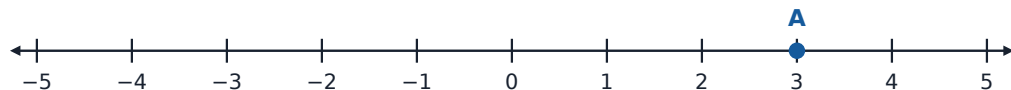
SET 29 • 6.NS.C.6.c

BACKWARD 3

Opposites on a number line • Grade 6 • The Number System • 6.NS.C.6.a

Point B (not shown) is the opposite of point A.

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q1/5

1087

QUESTION ID

S29-B3-Q1

TYPE

Short answer

CORRECT ANSWER

-3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q1/5

True or false?

–2.5 and 2.5 are the same distance from 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q2/5

1088

QUESTION ID

S29-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q2/5

What is the opposite of $1\frac{3}{4}$?

A. $-1\frac{3}{4}$

B. $-\frac{4}{7}$

C. $\frac{4}{7}$

D. $1\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q3/5

1089

QUESTION ID

S29-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-1\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q3/5

What is the opposite of -0.8 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q4/5

1090

QUESTION ID

S29-B3-Q4

TYPE

Short answer

CORRECT ANSWER

0.8

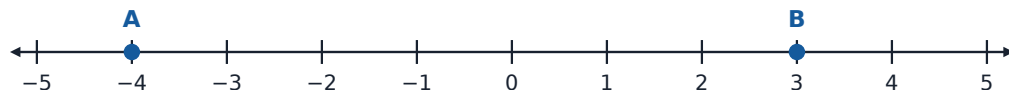
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q4/5

True or false?

Points A and B represent opposite numbers.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q5/5

1091

QUESTION ID

S29-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q5/5

SET 29 • 6.NS.C.6.c

FORWARD 1

Addition shown on a number line • Grade 7 • The Number System • 7.NS.A.1.b

Start at -2.5 on a number line and move 4 units to the right.

Where do you end?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q1/5

1093

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1.5

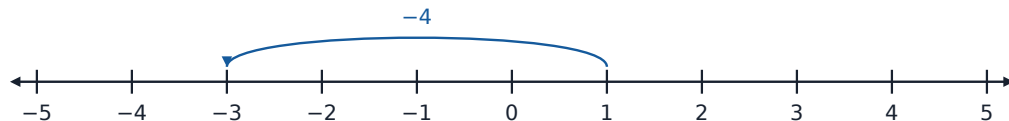
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q1/5

The number line shows $1 + (-4)$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q2/5

1094

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

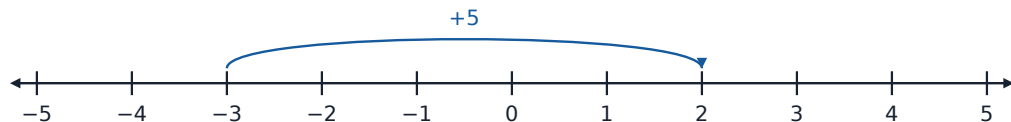
-3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q2/5

Which expression does the number line show?



A. $5 + 3$

B. $-3 - 5$

C. $-3 + 5$

D. $2 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q3/5

1095

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $-3 + 5$

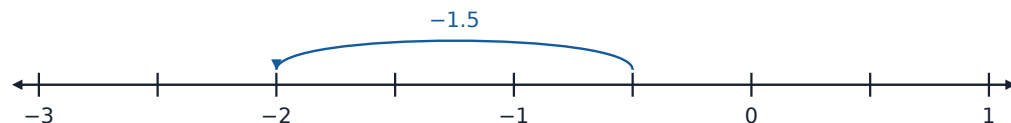
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q3/5

The number line shows $-0.5 + (-1.5)$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q4/5

1096

QUESTION ID

S29-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q4/5

True or false?

On a number line, $-3 + 7$ is 7 units to the right of -3 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q5/5

1097

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q5/5

SET 29 • 6.NS.C.6.c

FORWARD 2

Approximate irrational numbers on a number line • Grade 8 • The Number System • 8.NS.A.2

Between which two consecutive whole numbers is $\sqrt{20}$?

Between: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q1/5

1099

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q1/5

Which point best shows $\sqrt{10}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q2/5

1100

QUESTION ID

S29-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q2/5

Estimate $\sqrt{50}$ to the nearest tenth.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q3/5

1101

QUESTION ID

S29-F2-Q3

TYPE

Short answer

CORRECT ANSWER

7.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q3/5

Which list is ordered from least to greatest?

- A. 2.5, $\sqrt{5}$, $\sqrt{8}$
- B. $\sqrt{5}$, 2.5, $\sqrt{8}$
- C. $\sqrt{8}$, $\sqrt{5}$, 2.5
- D. $\sqrt{5}$, $\sqrt{8}$, 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q4/5

1102

QUESTION ID

S29-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $\sqrt{5}$, 2.5, $\sqrt{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q4/5

True or false?

π is between 3.1 and 3.2 on the number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q5/5

1103

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q5/5

Plot rational numbers on a horizontal or vertical number line

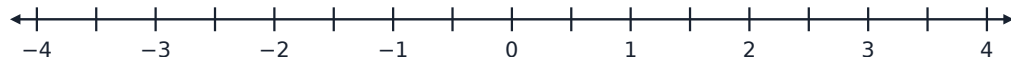
MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Placing fractions on a number line	3.NF.A.2
BACKWARD 2	Placing decimals on a number line	4.NF.C.6
BACKWARD 3	Opposites on a number line	6.NS.C.6.a
FORWARD 1	Represent addition on a number line	7.NS.A.1.b
FORWARD 2	Locate irrational numbers approximately on a number line	8.NS.A.2

SET 30 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label point A at -2.5 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q1/5

1106

QUESTION ID

S30-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

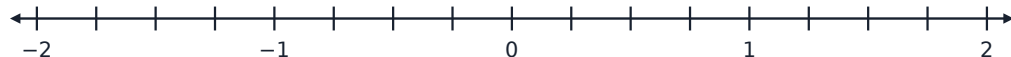
**Point A plotted at
 -2.5 (halfway
between -3 and -2)
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q1/5

Plot and label point B at $-\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q2/5

1107

QUESTION ID

S30-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Point B plotted at $-\frac{3}{4}$ (third tick to the left of 0) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

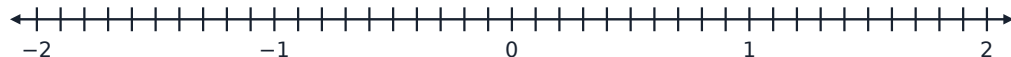
ID S30-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label -1.2 and its opposite on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q3/5

1108

QUESTION ID

S30-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at
 -1.2 and 1.2 and
labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

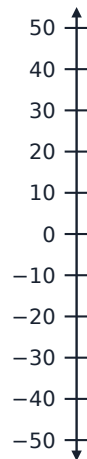
ID S30-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q3/5

1108

The number line shows elevations in feet.

Plot and label point C at -35 feet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q4/5

1109

QUESTION ID

S30-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

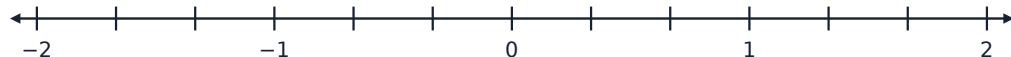
Point C plotted at -35 (halfway between -40 and -30 , below 0) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q4/5

Plot and label point D at $1\frac{2}{3}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q5/5

1110

QUESTION ID

S30-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Point D plotted at $1\frac{2}{3}$ (second tick to the right of 1) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

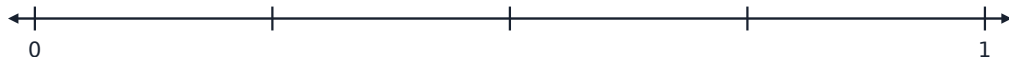
Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q5/5

SET 30 • 6.NS.C.6.c

BACKWARD 1

Placing fractions on a number line • Grade 3 • Number & Operations–Fractions
• 3.NF.A.2

Plot and label $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q1/5

1112

QUESTION ID

S30-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**$\frac{3}{4}$ plotted at the
third tick after 0 and
labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q1/5

Plot and label $\frac{2}{3}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q2/5

1113

QUESTION ID

S30-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

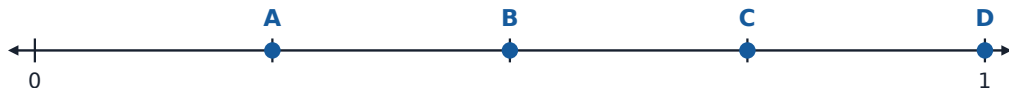
**$\frac{2}{3}$ plotted at the
second tick after 0
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q2/5

Which point is at $\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q3/5

1114

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

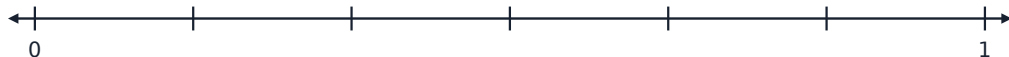
B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q3/5

Plot and label $\frac{5}{6}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q4/5

1115

QUESTION ID

S30-B1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**5/6 plotted at the
fifth tick after 0 and
labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q4/5

True or false?

To plot $\frac{3}{8}$, count 3 of the 8 equal parts from 0.



- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q5/5

1116

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 6.NS.C.6.c

BACKWARD 2

Placing decimals on a number line • Grade 4 • Number & Operations–Fractions
• 4.NF.C.6

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label 0.7 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q1/5

1118

QUESTION ID

S30-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**0.7 plotted at the
seventh tick after 0
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q1/5

1118

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label 2.4 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q2/5

1119

QUESTION ID

S30-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**2.4 plotted at the
fourth tick after 2
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q2/5

Which point is at 0.6?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q3/5

1120

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label 0.35 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q4/5

1121

QUESTION ID

S30-B2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**0.35 plotted at the
fifth tick after 0.3
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q4/5

1121

True or false?

On a number line marked in tenths, 0.9 is one tick mark to the left of 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q5/5

1122

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

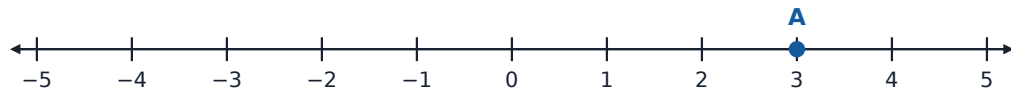
Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 6.NS.C.6.c

BACKWARD 3

Opposites on a number line • Grade 6 • The Number System • 6.NS.C.6.a

Point A is at 3. Plot and label its opposite.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q1/5

1124

QUESTION ID

S30-B3-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Point plotted at -3
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

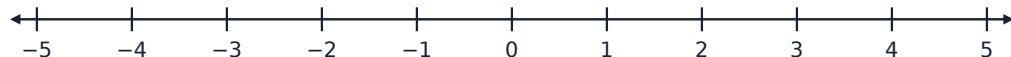
ID S30-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label 2 and -2 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q2/5

1125

QUESTION ID

S30-B3-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at 2
and -2 and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q2/5

1125

True or false?

A number and its opposite are the same distance from 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q3/5

1126

QUESTION ID

S30-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q3/5

Point B is at -1.5 . Where is its opposite?

- A.** At -1.5
- B.** At 1.5
- C.** At 0
- D.** At 0.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q4/5

1127

QUESTION ID

S30-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. At 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

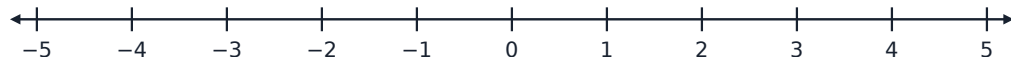
ID S30-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label the opposite of -4 .



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q5/5

1128

QUESTION ID

S30-B3-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**Point plotted at 4
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q5/5

1128

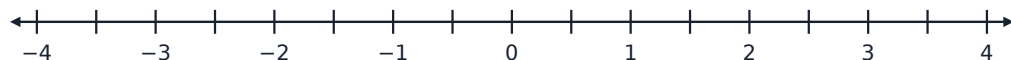
SET 30 • 6.NS.C.6.c

FORWARD 1

Represent addition on a number line • Grade 7 • The Number System •
7.NS.A.1.b

Draw an arrow on the number line to show $-2 + 3.5$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q1/5

1130

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE Arrow from -2 to 1.5

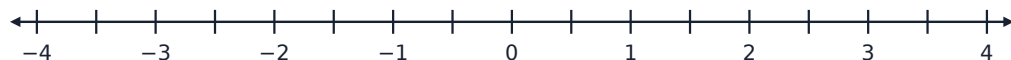
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q1/5

Draw arrows on the number line to show $1.5 + (-3)$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q2/5

1131

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-1.5

GRADING NOTE Arrows: 0 to 1.5, then 1.5 to -1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

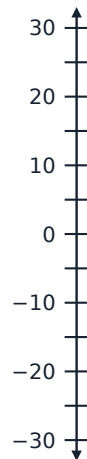
Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q2/5

Draw arrows on the vertical
number line to show

$$-10 + 25.$$

What is the sum?

Sum: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q3/5

1132

QUESTION ID

S30-F1-Q3

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Arrows: 0 down to -10 , then up 25 to 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q3/5

An arrow on a number line starts at $-\frac{1}{2}$ and moves $\frac{3}{4}$ unit to the right.

Where does it end?

A. $-1\frac{1}{4}$

B. $-\frac{1}{4}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q4/5

1133

QUESTION ID

S30-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q4/5

Draw arrows on the number line to show $-1\frac{1}{4} + (-\frac{3}{4})$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q5/5

1134

QUESTION ID

S30-F1-Q5

TYPE

Short answer

CORRECT ANSWER

-2

GRADING NOTE Arrows: 0 to $-1\frac{1}{4}$, then $-1\frac{1}{4}$ to -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q5/5

SET 30 • 6.NS.C.6.c

FORWARD 2

Locate irrational numbers approximately on a number line • Grade 8 • The Number System • 8.NS.A.2

Plot $\sqrt{10}$ at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q1/5

1136

QUESTION ID

S30-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Point just past 3, at
about 3.16 (between
3 and 3.5, closer to
3)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q1/5

Plot $\sqrt{2}$ at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q2/5

1137

QUESTION ID

S30-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Point at about 1.41
(between 1.4 and
1.5)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot π at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q3/5

1138

QUESTION ID

S30-F2-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Point at about 3.14
(between 3.1 and
3.2)**

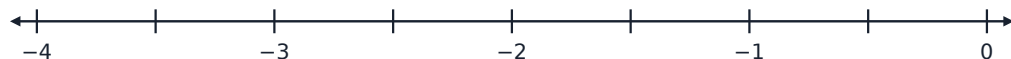
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q3/5

1138

Plot $-\sqrt{5}$ at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q4/5

1139

QUESTION ID

S30-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**Point at about -2.24
(between -2.5 and
 -2 , closer to -2)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot $\sqrt{30}$ at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q5/5

1140

QUESTION ID

S30-F2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**Point at about 5.48
(just to the left of
5.5)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q5/5

Name the coordinates of points in all four quadrants

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Ordered pairs and the axes	5.G.A.1
BACKWARD 2	Reading points in the first quadrant	5.G.A.2
BACKWARD 3	Quadrants and signs of coordinates	6.NS.C.6.b
FORWARD 1	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d
FORWARD 2	A function as a set of ordered pairs	8.F.A.1

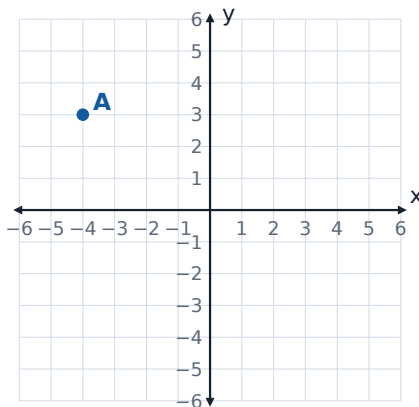
SET 31 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q1/5

1143

QUESTION ID

S31-M-Q1

TYPE

Short answer

CORRECT ANSWER

(-4, 3)

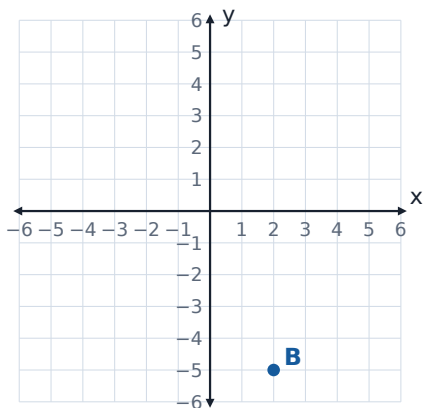
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q1/5

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q2/5

1144

QUESTION ID

S31-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, -5)

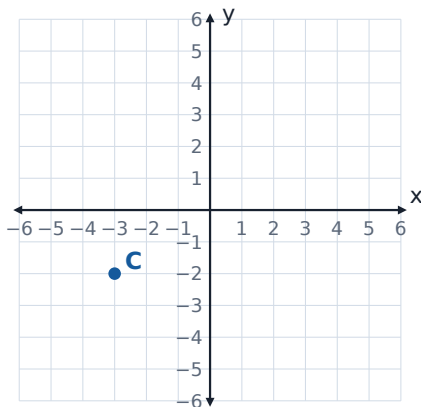
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q2/5

What are the coordinates of point C?

C = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q3/5

1145

QUESTION ID

S31-M-Q3

TYPE

Short answer

CORRECT ANSWER

(-3, -2)

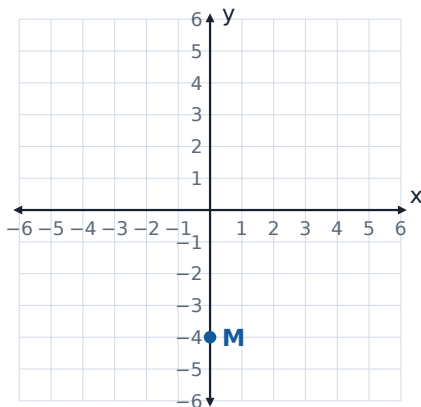
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q3/5

Which ordered pair names point M?

- A. $(0, 4)$
- B. $(4, 0)$
- C. $(0, -4)$
- D. $(-4, 0)$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q4/5

1146

QUESTION ID

S31-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(0, -4)$

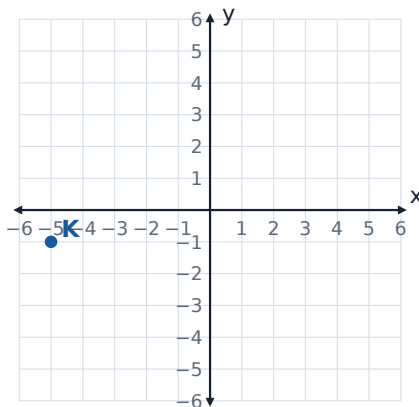
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q4/5

What are the coordinates of point K?

K = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q5/5

1147

QUESTION ID

S31-M-Q5

TYPE

Short answer

CORRECT ANSWER

(-5, -1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q5/5

SET 31 • 6.NS.C.6.c

BACKWARD 1

Ordered pairs and the axes • Grade 5 • Geometry • 5.G.A.1

True or false?

The x-coordinate tells how far to move from the origin along the x-axis.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q1/5

1149

QUESTION ID

S31-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q1/5

In the ordered pair (7, 3), what is the y-coordinate?

- A.** 3
- B.** 4
- C.** 7
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q2/5

1150

QUESTION ID

S31-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q2/5

The x-axis and the y-axis cross at a point.

What are its name and coordinates?

Name: _____

Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q3/5

1151

QUESTION ID

S31-B1-Q3

TYPE

Short answer

CORRECT ANSWER

**Name: the origin;
Coordinates: (0, 0)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q3/5

True or false?

(2, 5) and (5, 2) name the same point.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q4/5

1152

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q4/5

In an ordered pair, which number is written first?

- A.** The smaller number
- B.** The x-coordinate
- C.** The y-coordinate
- D.** The larger number

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q5/5

1153

QUESTION ID

S31-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. The x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q5/5

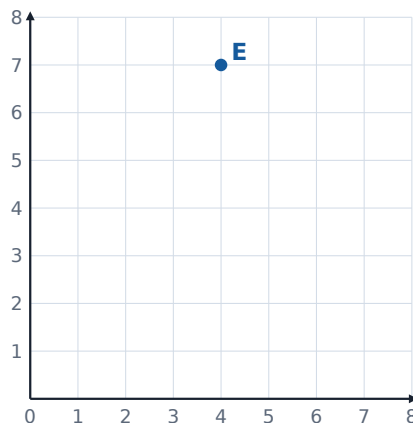
SET 31 • 6.NS.C.6.c

BACKWARD 2

Reading points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point E?

E = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

1155

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 7)

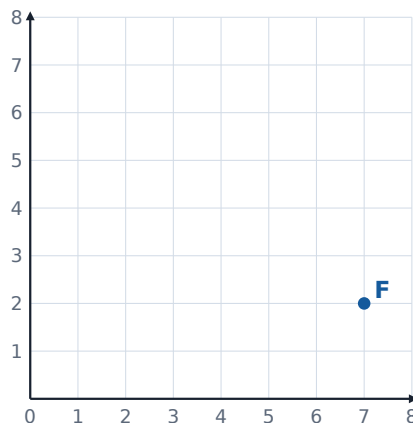
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

What are the coordinates of point F?

F = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

1156

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(7, 2)

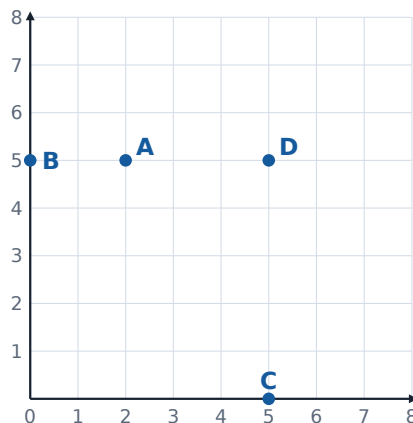
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

Which point is located at (0, 5)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

1157

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

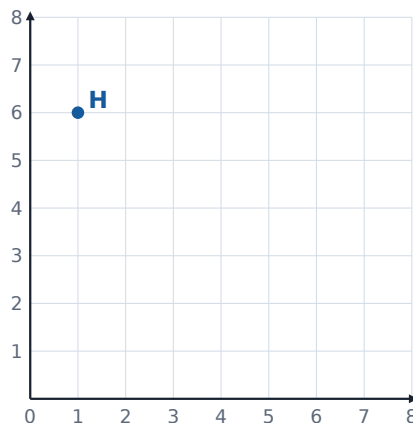
ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

True or false?

Point H is located at (6, 1).

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

1158

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

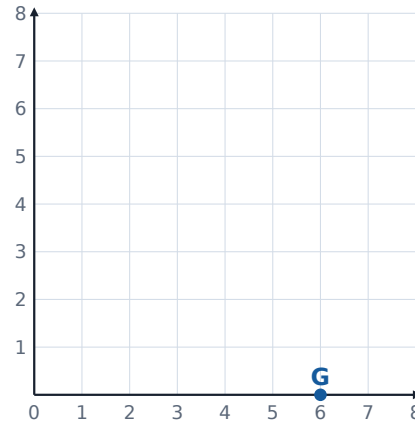
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

What are the coordinates of point G?

G = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

1159

QUESTION ID

S31-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(6, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 6.NS.C.6.c

BACKWARD 3

Quadrants and signs of coordinates • Grade 6 • The Number System •
6.NS.C.6.b

In which quadrant are both coordinates negative?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q1/5

1161

QUESTION ID

S31-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Quadrant III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q1/5

In which quadrant is $(5, -2)$?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q2/5

1162

QUESTION ID

S31-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q2/5

True or false?

$(-3, 4)$ is in Quadrant II.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q3/5

1163

QUESTION ID

S31-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q3/5

A point is in Quadrant IV.

Write "positive" or "negative" for each coordinate.

x-coordinate: _____

y-coordinate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q4/5

1164

QUESTION ID

S31-B3-Q4

TYPE

Short answer

CORRECT ANSWER

**x-coordinate:
positive;
y-coordinate:
negative**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q4/5

True or false?

The point $(0, -6)$ is in Quadrant III.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q5/5

1165

QUESTION ID

S31-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q5/5

SET 31 • 6.NS.C.6.c

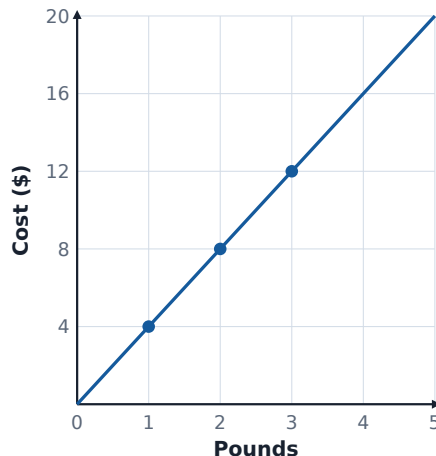
FORWARD 1

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of apples.

What does the point (2, 8) represent?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q1/5

1167

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**2 pounds of apples
cost \$8.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

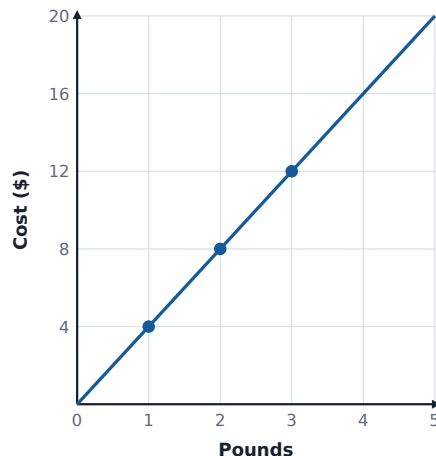
ID S31-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q1/5

The graph shows the cost of apples.

What does the point (1, 4) represent?

- A. 4 pounds cost \$1.
- B. 1 pound costs \$4.
- C. \$1 buys 4 pounds.
- D. The cost starts at \$4.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q2/5

1168

QUESTION ID

S31-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 1 pound costs \$4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q2/5

True or false?

On the graph of a proportional relationship between pounds and cost, the point (0, 0) means 0 pounds cost \$0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q3/5

1169

QUESTION ID

S31-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q3/5

The graph of a proportional relationship passes through (1, r) and (5, 30).

What is r ?

$r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q4/5

1170

QUESTION ID

S31-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q4/5

A graph shows the distance y in miles after x hours. It passes through $(3, 150)$.

What does this point mean?

- A.** In 3 hours, the distance is 150 miles.
- B.** The speed is 150 miles per hour.
- C.** In 150 hours, the distance is 3 miles.
- D.** The speed is 3 miles per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q5/5

1171

QUESTION ID

S31-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. In 3 hours, the distance is 150 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q5/5

SET 31 • 6.NS.C.6.c

FORWARD 2

A function as a set of ordered pairs • Grade 8 • Functions • 8.F.A.1

True or false?

The set of ordered pairs (1, 2), (2, 4), (3, 6), (4, 8) represents a function.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q1/5

1173

QUESTION ID

S31-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q1/5

Which set of ordered pairs does NOT represent a function?

- A. $(-1, 3), (0, 5), (1, 7)$
- B. $(1, 4), (2, 4), (3, 4)$
- C. $(0, 0), (1, 1), (2, 2)$
- D. $(2, 5), (2, 7), (3, 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q2/5

1174

QUESTION ID

S31-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $(2, 5), (2, 7), (3, 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q2/5

Does the set $(4, 1), (5, 2), (4, 3)$ represent a function?
Write yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q3/5

1175

QUESTION ID

S31-F2-Q3

TYPE

Short answer

CORRECT ANSWER

No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

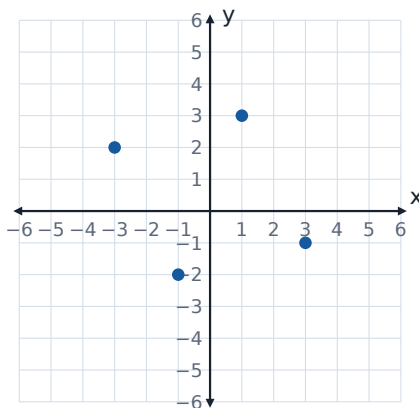
ID S31-F2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q3/5

True or false?

The graph represents a function.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q4/5

1176

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q4/5

A function assigns to each input ____.

- A.** at least two outputs
- B.** no output
- C.** the same output as the input
- D.** exactly one output

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q5/5

1177

QUESTION ID

S31-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

**D. exactly one
output**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q5/5

Plot points in all four quadrants

MAIN

Grade 6 main question

6.NS.C.6.c

BACKWARD 1

Which direction to move first

5.G.A.1

BACKWARD 2

Plotting points in the first quadrant

5.G.A.2

BACKWARD 3

Direction from the sign of a coordinate

6.NS.C.6.b

FORWARD 1

Graph pairs to decide whether a relationship is proportional

7.RP.A.2.a

FORWARD 2

Graph proportional relationships

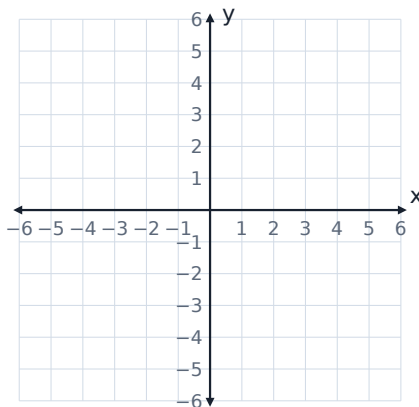
8.EE.B.5

SET 32 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label point $P(-3, 4)$
and point $Q(2, -5)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q1/5

1180

QUESTION ID

S32-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

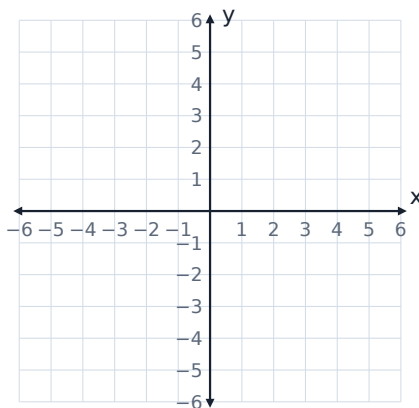
**P plotted at $(-3, 4)$
and Q plotted at $(2, -5)$, both labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q1/5

Plot and label point R($-5, -2$) and point S($4, 1$).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q2/5

1181

QUESTION ID

S32-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

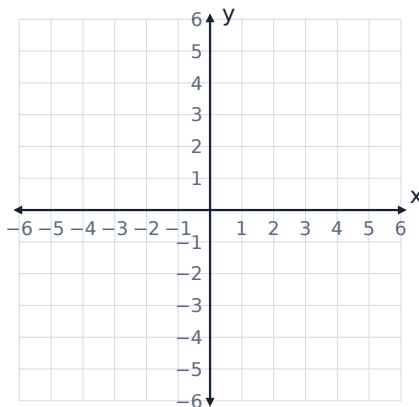
**R plotted at $(-5, -2)$
and S plotted at $(4, 1)$, both labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q2/5

Plot and label point $T(0, -3)$
and point $U(-4, 0)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q3/5

1182

QUESTION ID

S32-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

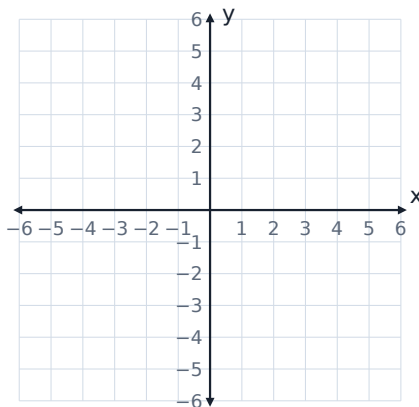
**T plotted at $(0, -3)$
on the y-axis and U
plotted at $(-4, 0)$ on
the x-axis, both
labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q3/5

Plot and label point $V(-2.5, 3)$ and point $W(1.5, -2)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q4/5

1183

QUESTION ID

S32-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

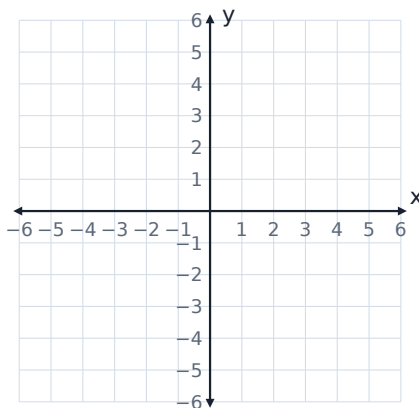
V plotted at $(-2.5, 3)$ and W plotted at $(1.5, -2)$, both labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q4/5

Plot and label point $X(-6, -6)$
and point $Y(5, -4)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q5/5

1184

QUESTION ID

S32-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**X plotted at $(-6, -6)$
and Y plotted at $(5, -4)$, both labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q5/5

SET 32 • 6.NS.C.6.c

BACKWARD 1

Which direction to move first • Grade 5 • Geometry • 5.G.A.1

True or false?

To plot (6, 1), first move 6 units along the x-axis.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q1/5

1186

QUESTION ID

S32-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q1/5

To plot (2, 7) from the origin, what do you do first?

- A. Move 7 units up
- B. Move 2 units right
- C. Move 2 units up
- D. Move 7 units right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q2/5

1187

QUESTION ID

S32-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Move 2 units right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q2/5

True or false?

To plot (0, 5), you do not move left or right.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q3/5

1188

QUESTION ID

S32-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q3/5

To plot (4, 3), how many units up do you move?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q4/5

1189

QUESTION ID

S32-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q4/5

Which ordered pair is 3 units right and 6 units up from the origin?

- A. (3, 3)
- B. (3, 6)
- C. (6, 3)
- D. (6, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q5/5

1190

QUESTION ID

S32-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

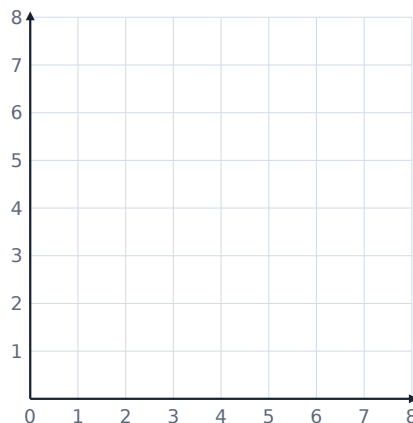
Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 6.NS.C.6.c

BACKWARD 2

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

Plot and label point A at (2, 6).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q1/5

1192

QUESTION ID

S32-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

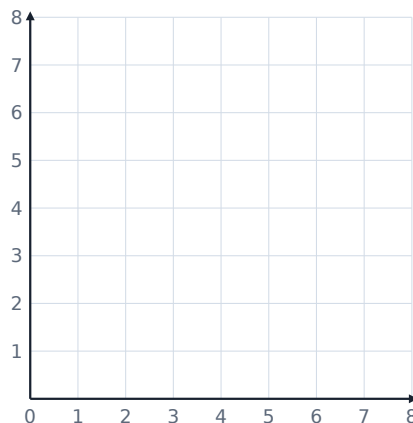
Point A plotted at (2, 6) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q1/5

Plot and label point B at (5, 1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q2/5

1193

QUESTION ID

S32-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

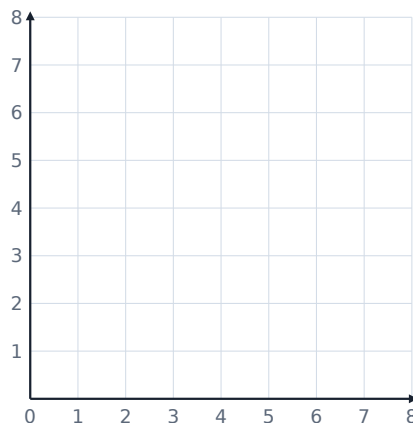
Point B plotted at (5, 1) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q2/5

Plot and label point C at (0, 7).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q3/5

1194

QUESTION ID

S32-B2-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

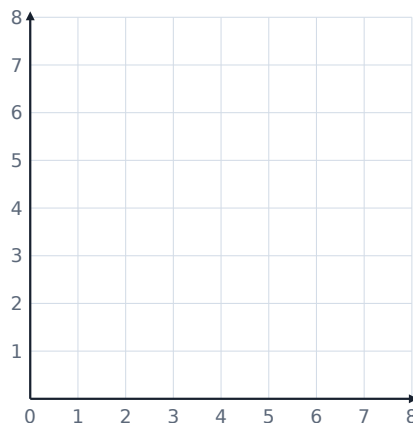
Point C plotted at (0, 7) on the y-axis and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q3/5

Plot and label point D at (7, 7).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q4/5

1195

QUESTION ID

S32-B2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

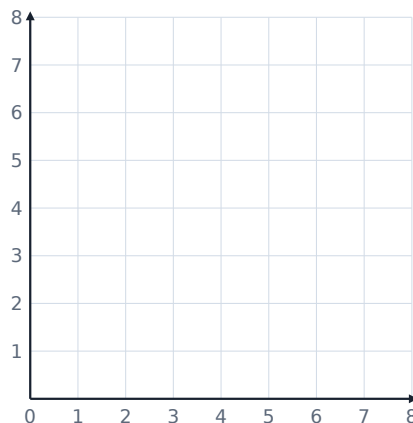
Point D plotted at (7, 7) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q4/5

Plot and label point E at (5, 0).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q5/5

1196

QUESTION ID

S32-B2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Point E plotted at (5, 0) on the x-axis and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 6.NS.C.6.c

BACKWARD 3

Direction from the sign of a coordinate • Grade 6 • The Number System •
6.NS.C.6.b

True or false?

To plot a point with a negative x-coordinate, you move left from the origin.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q1/5

1198

QUESTION ID

S32-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q1/5

To plot $(-2, -5)$, which directions do you move?

- A. Right 2, down 5
- B. Right 2, up 5
- C. Left 2, down 5
- D. Left 5, down 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q2/5

1199

QUESTION ID

S32-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Left 2, down 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q2/5

True or false?

To plot $(3, -4)$, you move up 4 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q3/5

1200

QUESTION ID

S32-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

To plot $(-6, 1)$, do you move left or right?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q4/5

1201

QUESTION ID

S32-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q4/5

1201

Which point is left of the y-axis and above the x-axis?

- A.** $(-3, 2)$
- B.** $(3, -2)$
- C.** $(3, 2)$
- D.** $(-3, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q5/5

1202

QUESTION ID

S32-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $(-3, 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q5/5

SET 32 • 6.NS.C.6.c

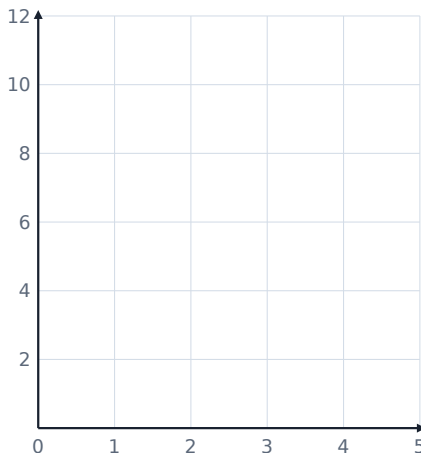
FORWARD 1

Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs (1, 3), (2, 6), and (3, 9).

Do they lie on a straight line through the origin?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q1/5

1204

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Points (1, 3), (2, 6), (3, 9) graphed; they lie on a line through (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

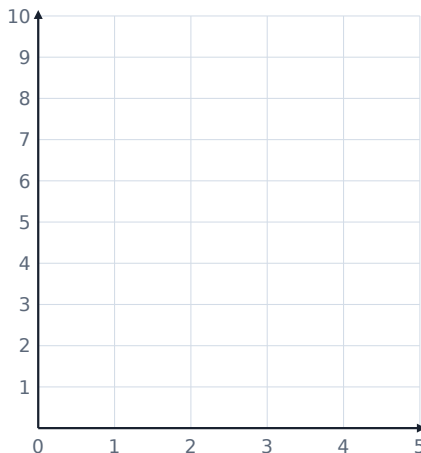
ID S32-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q1/5

Graph the pairs (1, 3), (2, 5), and (3, 7).

Is the relationship proportional?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q2/5

1205

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Points (1, 3), (2, 5), (3, 7) graphed; the line does not pass through (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

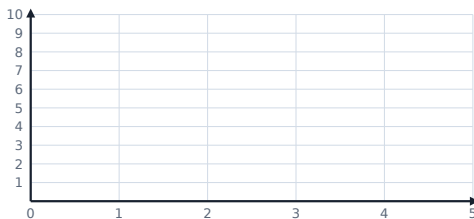
ID S32-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q2/5

Graph the pairs from the table.

Is the relationship proportional?

x	1	2	4
y	2.5	5	10



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q3/5

1206

QUESTION ID

S32-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Points (1, 2.5), (2, 5), (4, 10) graphed; they lie on a line through (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q3/5

True or false?

The points (2, 1), (4, 2), and (6, 3) lie on a line through the origin, so the relationship is proportional.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q4/5

1207

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q4/5

Which set of pairs would graph as a proportional relationship?

- A. (1, 1), (2, 4), (3, 9)
- B. (1, 3), (2, 4), (3, 5)
- C. (0, 2), (1, 4), (2, 6)
- D. (2, 3), (4, 6), (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q5/5

1208

QUESTION ID

S32-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 3), (4, 6), (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q5/5

SET 32 • 6.NS.C.6.c

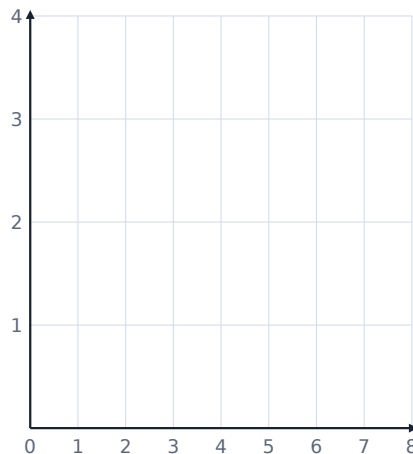
FORWARD 2

Graph proportional relationships • Grade 8 • Expressions & Equations •
8.EE.B.5

Graph $y = 0.5x$ on the coordinate plane.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q1/5

1210

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.5

GRADING NOTE Graph: line through (0, 0), (2, 1), (4, 2), (8, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

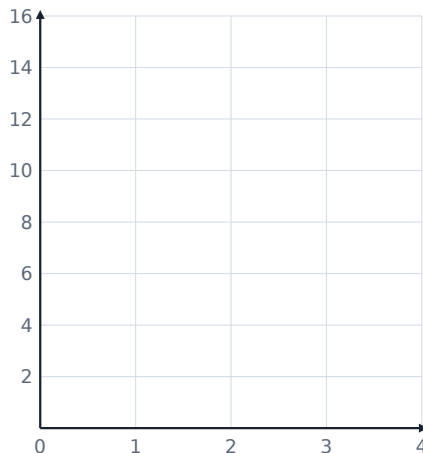
ID S32-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q1/5

Graph $y = 4x$ on the coordinate plane.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q2/5

1211

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Graph: line through (0, 0), (1, 4), (4, 16)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

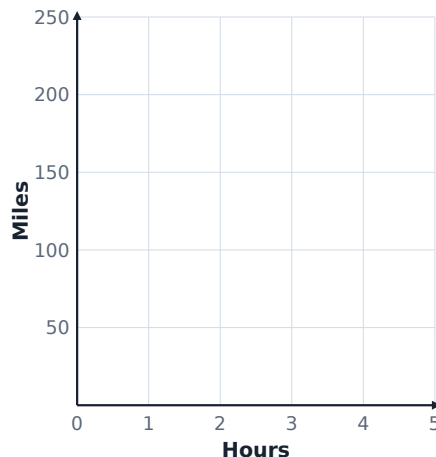
ID S32-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q2/5

A car travels 50 miles per hour. Graph the distance y after x hours.

What is the slope?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q3/5

1212

QUESTION ID

S32-F2-Q3

TYPE

Short answer

CORRECT ANSWER

50

GRADING NOTE Graph: line through (0, 0), (1, 50), (5, 250)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q3/5

Which point is on the graph of $y = 3x$?

- A. (3, 1)
- B. (12, 4)
- C. (4, 12)
- D. (4, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q4/5

1213

QUESTION ID

S32-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (4, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q4/5

True or false?

The graph of $y = \frac{2}{3}x$ passes through (3, 2).

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q5/5

1214

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q5/5

Interpret inequalities as relative position on a number line

MAIN	Grade 6 main question
BACKWARD 1	Comparison symbols
BACKWARD 2	Position on a number line
BACKWARD 3	Negative numbers on a number line
FORWARD 1	Solve and graph inequalities
FORWARD 2	Compare and order irrational numbers

6.NS.C.7.a

2.NBT.A.4

2.MD.B.6

6.NS.C.6.c

7.EE.B.4.b

8.NS.A.2

SET 33 • 6.NS.C.7.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.a

$$-4 > -10$$

Which statement explains this inequality?

- A. -4 is to the left of -10 on a number line.
- B. 4 is less than 10 .
- C. -4 is farther from 0 than -10 .
- D. -4 is to the right of -10 on a number line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q1/5

1217

QUESTION ID

S33-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. -4 is to the right of -10 on a number line.

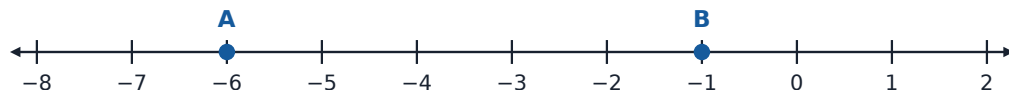
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q1/5

True or false?

The number line shows that $-6 > -1$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q2/5

1218

QUESTION ID

S33-M-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q2/5

Write an inequality that compares -4 and -9 .

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q3/5

1219

QUESTION ID

S33-M-Q3

TYPE

Short answer

CORRECT ANSWER

$-4 > -9$

GRADING NOTE $-9 < -4$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q3/5

Which statement is true?

- A.** $-5 < -2$
- B.** $-2 < -5$
- C.** $-5 > 0$
- D.** $-5 > -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q4/5

1220

QUESTION ID

S33-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

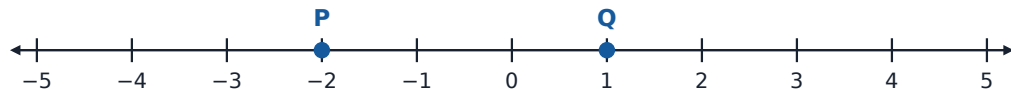
A. $-5 < -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q4/5

Write an inequality using the numbers at points P and Q.



Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q5/5

1221

QUESTION ID

S33-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-2 < 1$

GRADING NOTE $1 > -2$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q5/5

SET 33 • 6.NS.C.7.a

BACKWARD 1

Comparison symbols • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $<$, or $=$ to compare.

452 ____ 425

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q1/5

1223

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

 $>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q1/5

1223

True or false?

$$67 < 76$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q2/5

1224

QUESTION ID

S33-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q2/5

Which statement is true?

- A.** $309 < 290$
- B.** $309 = 290$
- C.** $309 > 290$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q3/5

1225

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $309 > 290$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q3/5

Write $>$, $<$, or $=$ to compare.

138 ____ 183

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q4/5

1226

QUESTION ID

S33-B1-Q4

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q4/5

True or false?

500 > 505

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q5/5

1227

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 6.NS.C.7.a

BACKWARD 2

Position on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

True or false?

On a number line, 8 is to the right of 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q1/5

1229

QUESTION ID

S33-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q1/5

Which number is farthest to the left on a number line?

- A.** 2
- B.** 5
- C.** 7
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q2/5

1230

QUESTION ID

S33-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q2/5

Which point is farther to the left, A or B?



Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q3/5

1231

QUESTION ID

S33-B2-Q3

TYPE

Short answer

CORRECT ANSWER

B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q3/5

True or false?

Numbers get greater as you move to the left on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q4/5

1232

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q4/5

Which number is to the right of 14 on a number line?

- A. 9
- B. 11
- C. 14
- D. 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q5/5

1233

QUESTION ID

S33-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 6.NS.C.7.a

BACKWARD 3

Negative numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q1/5

1235

QUESTION ID

S33-B3-Q1

TYPE

Short answer

CORRECT ANSWER

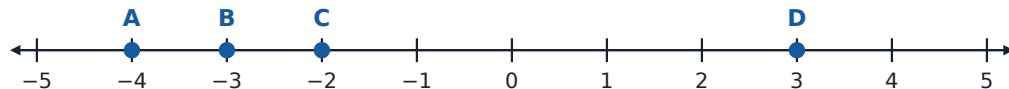
-6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q1/5

Which point is at -3 ?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q2/5

1236

QUESTION ID

S33-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q2/5

True or false?

–4 is to the left of –1 on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q3/5

1237

QUESTION ID

S33-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q3/5

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q4/5

1238

QUESTION ID

S33-B3-Q4

TYPE

Short answer

CORRECT ANSWER

-8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q4/5

Which point is at -5 ?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q5/5

1239

QUESTION ID

S33-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q5

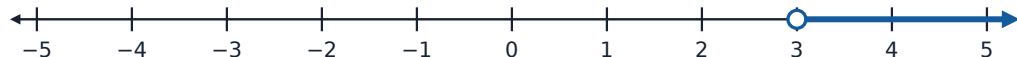
Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q5/5

SET 33 • 6.NS.C.7.a

FORWARD 1

Solve and graph inequalities • Grade 7 • Expressions & Equations • 7.EE.B.4.b

Which inequality does the graph show?



A. $x > 3$

B. $x \leq 3$

C. $x < 3$

D. $x \geq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q1/5

1241

QUESTION ID

S33-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $x > 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q1/5

Solve the inequality.

$$2x + 5 > 11$$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q2/5

1242

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$$x > 3$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q2/5

Solve the inequality.

$$-3x + 4 \leq 16$$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q3/5

1243

QUESTION ID

S33-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$$x \geq -4$$

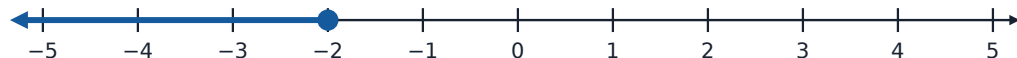
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q3/5

True or false?

The graph shows the solutions of $4x - 1 \leq -9$.



- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q4/5

1244

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q4/5

Solve the inequality.

$$5x - 2 < 18$$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q5/5

1245

QUESTION ID

S33-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$$x < 4$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q5/5

SET 33 • 6.NS.C.7.a

FORWARD 2

Compare and order irrational numbers • Grade 8 • The Number System •
8.NS.A.2

Which is greater, $\sqrt{50}$ or 7.2?

Greater: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q1/5

1247

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7.2

GRADING NOTE $\sqrt{50} \approx 7.07$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q1/5

Which number is between 3 and 4?

- A. $\sqrt{12}$
- B. $\sqrt{8}$
- C. $\sqrt{3}$
- D. $\sqrt{17}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q2/5

1248

QUESTION ID

S33-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $\sqrt{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q2/5

True or false?

$$\sqrt{30} < 5.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q3/5

1249

QUESTION ID

S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q3/5

Order from least to greatest.

π , $\sqrt{8}$, 3.1

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q4/5

1250

QUESTION ID

S33-F2-Q4

TYPE

Short answer

CORRECT ANSWER

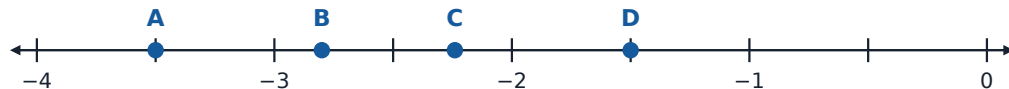
$\sqrt{8}$, 3.1, π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q4/5

Which point best shows $-\sqrt{5}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q5/5

1251

QUESTION ID

S33-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q5/5

Write, interpret, and explain order statements in real-world contexts

MAIN	Grade 6 main question	6.NS.C.7.b
BACKWARD 1	Negative numbers in context	6.NS.C.5
BACKWARD 2	Inequality symbols	1.NBT.B.3
BACKWARD 3	Order as position on a number line	6.NS.C.7.a
FORWARD 1	Write and solve inequalities for real-world situations	7.EE.B.4.b
FORWARD 2	Compare quantities written with powers of 10	8.EE.A.3

SET 34 • 6.NS.C.7.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.b

A temperature of -2°F is warmer than a temperature of -11°F .

Write an inequality to show this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q1/5

1254

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-2 > -11$

GRADING NOTE $-11 < -2$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q1/5

Account A has a balance of $-\$25$. Account B has a balance of $-\$40$.

Which statement is true?

- A.** $-25 > -40$, so Account A has the greater balance.
- B.** $-40 > -25$, so Account B has the greater balance.
- C.** $-25 < -40$, so Account A owes more.
- D.** The balances are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q2/5

1255

QUESTION ID

S34-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $-25 > -40$, so Account A has the greater balance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q2/5

A fish swims at -12 feet. A turtle swims at -5 feet.

Write an inequality to compare the positions. Which animal is deeper?

Inequality: _____

Deeper animal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q3/5

1256

QUESTION ID

S34-M-Q3

TYPE

Short answer

CORRECT ANSWER

**Inequality: $-12 < -5$
(or $-5 > -12$);
Deeper animal: the
fish**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q3/5

True or false?

$-15 < -10$ means that -15°F is colder than -10°F .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q4/5

1257

QUESTION ID

S34-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q4/5

The lowest point in a valley is -86 meters. A beach is at 0 meters.

Which statement is true?

- A.** $-86 < 0$, so the valley's lowest point is lower. **B.** $-86 = 0$
- C.** $0 < -86$, so the beach is lower. **D.** $-86 > 0$, so the valley's lowest point is higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q5/5

1258

QUESTION ID

S34-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $-86 < 0$, so the valley's lowest point is lower.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q5/5

SET 34 • 6.NS.C.7.b

BACKWARD 1

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 12 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q1/5

1260

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

-12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q1/5

True or false?

A debt of \$30 can be written as -30 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q2/5

1261

QUESTION ID

S34-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q2/5

Which situation could be represented by -6 ?

- A.** A loss of 6 yards
- B.** A deposit of \$6
- C.** A gain of 6 yards
- D.** 6 feet above sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q3/5

1262

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A loss of 6 yards

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q3/5

Write an integer for 50 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q4/5

1263

QUESTION ID

S34-B1-Q4

TYPE

Short answer

CORRECT ANSWER

−50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q4/5

True or false?

A temperature of 4 degrees above zero is written as -4 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q5/5

1264

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 6.NS.C.7.b

BACKWARD 2

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

42 ____ 24

A. <

B. >

C. =

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q1/5

1266

QUESTION ID

S34-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. >

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q1/5

True or false?

15 > 51

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q2/5

1267

QUESTION ID

S34-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $<$, or $=$ to compare.

73 ____ 37

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q3/5

1268

QUESTION ID

S34-B2-Q3

TYPE

Short answer

CORRECT ANSWER

 $>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q3/5

1268

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q4/5

1269

QUESTION ID

S34-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q4/5

True or false?

89 < 98

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q5/5

1270

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • 6.NS.C.7.b

BACKWARD 3

Order as position on a number line • Grade 6 • The Number System •
6.NS.C.7.a

True or false?

$-2 > -6$ because -2 is to the right of -6 on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q1/5

1272

QUESTION ID

S34-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q1/5

Which number is greatest?

- A.** -9
- B.** -7
- C.** -4
- D.** -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q2/5

1273

QUESTION ID

S34-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q2/5

–5 is to the left of 3 on a number line.

Write an inequality to show this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q3/5

1274

QUESTION ID

S34-B3-Q3

TYPE

Short answer

CORRECT ANSWER

–5 < 3

GRADING NOTE $3 > -5$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q3/5

True or false?

–8 is to the right of –3 on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q4/5

1275

QUESTION ID

S34-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q4/5

Which number is to the left of -4 on a number line?

- A.** -6
- B.** -2
- C.** 0
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q5/5

1276

QUESTION ID

S34-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. -6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q5/5

SET 34 • 6.NS.C.7.b

FORWARD 1

Write and solve inequalities for real-world situations • Grade 7 • Expressions & Equations • 7.EE.B.4.b

A submarine at -40 m descends 12 m per minute. It must stay above -160 m.

Write and solve an inequality for the number of minutes t it can descend.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q1/5

1278

QUESTION ID

S34-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**$-40 - 12t > -160$,
so $t < 10$ (fewer than
10 minutes)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q1/5

You have \$25 and spend \$4 per day. You want to keep at least \$5.

Write and solve an inequality for the number of days d .

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q2/5

1279

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

**$25 - 4d \geq 5$, so $d \leq 5$
(at most 5 days)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q2/5

A freezer is at 2°C . It cools 3°C per hour.

For which numbers of hours h is its temperature below -10°C ?

- A.** $h > 4$ **B.** $h > -4$
C. $h < -4$ **D.** $h < 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q3/5

1280

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $h > 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q3/5

A diver at -8 ft descends 5 ft per minute.

Write and solve an inequality to find when the diver is deeper than -48 ft.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q4/5

1281

QUESTION ID

S34-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**$-8 - 5t < -48$, so
 $t > 8$ (after more
than 8 minutes)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q4/5

True or false?

The solution of $-2x + 1 > 9$ is $x < -4$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q5/5

1282

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q5/5

SET 34 • 6.NS.C.7.b

FORWARD 2

Compare quantities written with powers of 10 • Grade 8 • Expressions & Equations • 8.EE.A.3

A mountain is about 9×10^3 m tall. An ocean trench is about -1.1×10^4 m deep.

Which is farther from sea level?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q1/5

1284

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The ocean trench

GRADING NOTE $|-1.1 \times 10^4| = 11,000 \text{ m} > 9,000 \text{ m}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q1/5

Which number is greater?

A. 3×10^{-2}

B. 8×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q2/5

1285

QUESTION ID

S34-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3×10^{-2}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q2/5

A city has about 2×10^6 people. A town has about 5×10^4 people.

How many times as many people live in the city?

Times as many: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q3/5

1286

QUESTION ID

S34-F2-Q3

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q3/5

Order from least to greatest.

$$4 \times 10^3, 9 \times 10^2, 1 \times 10^4$$

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q4/5

1287

QUESTION ID

S34-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**$9 \times 10^2, 4 \times 10^3,$
 1×10^4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q4/5

True or false?

$$-5 \times 10^3 < -2 \times 10^3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q5/5

1288

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q5/5

Find absolute value as distance from 0

MAIN	Grade 6 main question	6.NS.C.7.c
BACKWARD 1	Distance on a number line	2.MD.B.6
BACKWARD 2	Opposite numbers	6.NS.C.6.a
FORWARD 1	Distance between rational numbers as $ p - q $	7.NS.A.1.c
FORWARD 2	Square roots and cube roots as solutions	8.EE.A.2

SET 35 • 6.NS.C.7.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-18|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q1/5

1291

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q1/5

Which number has the greatest absolute value?

- A.** -12
- B.** -5
- C.** 8
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q2/5

1292

QUESTION ID

S35-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. -12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q2/5

How far is -7.5 from 0 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q3/5

1293

QUESTION ID

S35-M-Q3

TYPE

Short answer

CORRECT ANSWER

7.5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q3/5

True or false?

$$|-6| = |6|$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q4/5

1294

QUESTION ID

S35-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q4/5

Which two numbers have an absolute value of 11?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q5/5

1295

QUESTION ID

S35-M-Q5

TYPE

Short answer

CORRECT ANSWER

11 and -11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q5/5

SET 35 • 6.NS.C.7.c

BACKWARD 1

Distance on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How many units apart are points A and B?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q1/5

1297

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q1/5

True or false?

The distance from 0 to 5 is the same as the distance from 5 to 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q2/5

1298

QUESTION ID

S35-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q2/5

How many units from 0 is the number 11?

- A.** 0
- B.** 1
- C.** 10
- D.** 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q3/5

1299

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q3/5

What is the distance from 4 to 10 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q4/5

1300

QUESTION ID

S35-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q4/5

How many units is point C from 0?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q5/5

1301

QUESTION ID

S35-B1-Q5

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 6.NS.C.7.c

BACKWARD 2

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 15?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q1/5

1303

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

-15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q1/5

True or false?

Opposite numbers are the same distance from 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q2/5

1304

QUESTION ID

S35-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q2/5

Which number is the opposite of $-\frac{2}{3}$?

A. $-\frac{3}{2}$

B. $-\frac{2}{3}$

C. $\frac{2}{3}$

D. $\frac{3}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q3/5

1305

QUESTION ID

S35-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q3/5

What is the opposite of 0?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q4/5

1306

QUESTION ID

S35-B2-Q4

TYPE

Short answer

CORRECT ANSWER

0

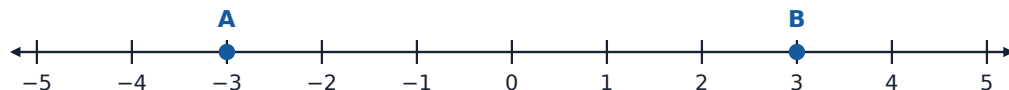
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q4/5

True or false?

Points A and B represent opposite numbers.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q5/5

1307

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 6.NS.C.7.c

FORWARD 1

Distance between rational numbers as $|p - q|$ • Grade 7 • The Number System
• 7.NS.A.1.c

Find the distance between -4 and 7 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q1/5

1309

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q1/5

Find the value.

$$|-3.5 - 2|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q2/5

1310

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q2/5

Which expression gives the distance between -9 and -2 ?

- A. $-9 - 2$
- B. $|-9 - (-2)|$
- C. $|-9 + (-2)|$
- D. $-9 + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q3/5

1311

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $|-9 - (-2)|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q3/5

True or false?

The distance between -6 and 6 is 0 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q4/5

1312

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q4/5

Find the distance between $-2\frac{1}{2}$ and $1\frac{1}{4}$ on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q5/5

1313

QUESTION ID

S35-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q5/5

SET 35 • 6.NS.C.7.c

FORWARD 2

Square roots and cube roots as solutions • Grade 8 • Expressions & Equations •
8.EE.A.2

Solve.

$$x^2 = 49$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q1/5

1315

QUESTION ID

S35-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 7$ or $x = -7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q1/5

Solve.

$$x^3 = 64$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q2/5

1316

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q2/5

What are the solutions of $x^2 = 81$?

- A. $x = -9$ only
- B. $x = 9$ only
- C. $x = 40.5$
- D. $x = 9$ or $x = -9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q3/5

1317

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = 9$ or $x = -9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q3/5

Solve. Write the answer using a square root symbol.

$$x^2 = 20$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q4/5

1318

QUESTION ID

S35-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$$x = \sqrt{20} \text{ or } x = -\sqrt{20}$$

GRADING NOTE $\pm\sqrt{20}$; also accept $\pm 2\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q4/5

True or false?

Both 6 and -6 are solutions of $x^2 = 36$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q5/5

1319

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q5/5

Interpret absolute value as magnitude in a real-world situation

MAIN	Grade 6 main question	6.NS.C.7.c
BACKWARD 1	Negative numbers in context	6.NS.C.5
BACKWARD 2	Distance from 0	2.MD.B.6
BACKWARD 3	Opposites are the same distance from 0	6.NS.C.6.a
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Compare magnitudes with powers of 10	8.EE.A.3

SET 36 • 6.NS.C.7.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.c

An account balance is $-\$45$.

What is the absolute value of the balance, and what does it describe?

Absolute value: _____

It describes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q1/5

1322

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

**Absolute value: 45;
It describes: a debt
of \$45 (the amount
owed)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q1/5

A diver is at an elevation of -42 feet.

How far is the diver from sea level?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q2/5

1323

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

42 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q2/5

A stock's value changed by $-\$8.50$ today.

What is the size of the change?

- A. $-\$8.50$
- B. $\$0$
- C. $\$8.50$
- D. $\$17.00$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q3/5

1324

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $\$8.50$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q3/5

The temperature is -13°F .

How many degrees below zero is it?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q4/5

1325

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

13 degrees

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q4/5

A submarine is at -275 m. A plane is at 275 m.

Which is farther from sea level? Explain using absolute value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q5/5

1326

QUESTION ID

S36-M-Q5

TYPE

Short answer

CORRECT ANSWER

**Neither: both are
275 m from sea
level, because
 $|-275| = |275|$
 $= 275$.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q5/5

SET 36 • 6.NS.C.7.c

BACKWARD 1

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

True or false?

–30 can represent owing \$30.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q1/5

1328

QUESTION ID

S36-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q1/5

Write an integer for 42 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q2/5

1329

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

-42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q2/5

Which situation is represented by +25?

- A.** A loss of 25 yards
- B.** 25 feet below sea level
- C.** A debt of \$25
- D.** A deposit of \$25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q3/5

1330

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A deposit of \$25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q3/5

Write an integer for a debt of \$18.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q4/5

1331

QUESTION ID

S36-B1-Q4

TYPE

Short answer

CORRECT ANSWER

−18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q4/5

True or false?

A temperature of 0°F means there is no temperature.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q5/5

1332

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 6.NS.C.7.c

BACKWARD 2

Distance from 0 • Grade 2 • Measurement & Data • 2.MD.B.6

How many units from 0 is point A?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q1/5

1334

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q1/5

True or false?

The distance from 0 to 13 is 13 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q2/5

1335

QUESTION ID

S36-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q2/5

Which number is 5 units from 0 on this number line?



A. 0

B. 5

C. 10

D. 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q3/5

1336

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q3/5

How far is 20 from 0 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q4/5

1337

QUESTION ID

S36-B2-Q4

TYPE

Short answer

CORRECT ANSWER

20 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q4/5

True or false?

The number that is 0 units from 0 is 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q5/5

1338

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 6.NS.C.7.c

BACKWARD 3

Opposites are the same distance from 0 • Grade 6 • The Number System •
6.NS.C.6.a

True or false?

–42 and 42 are the same distance from 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q1/5

1340

QUESTION ID

S36-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q1/5

What is the opposite of -275 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q2/5

1341

QUESTION ID

S36-B3-Q2

TYPE

Short answer

CORRECT ANSWER

275

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q2/5

Which number is the same distance from 0 as -8.5 ?

- A.** -5.8
- B.** 0
- C.** 8.5
- D.** 85

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q3/5

1342

QUESTION ID

S36-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q3/5

True or false?

The opposite of 13 is -13 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q4/5

1343

QUESTION ID

S36-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q4/5

Name the number that is the same distance from 0 as -45 , on the other side of 0.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q5/5

1344

QUESTION ID

S36-B3-Q5

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q5/5

SET 36 • 6.NS.C.7.c

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A company lost \$2.5 million in one year and earned a profit of \$1.8 million the next year.

What is the total for the two years?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q1/5

1346

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−\$0.7 million

GRADING NOTE A loss of \$700,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q1/5

The temperature was -6°F . It dropped 11°F .

What is the new temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q2/5

1347

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-17°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q2/5

A diver at -35.5 feet rises 12.25 feet.

What is the diver's new position?

- A. -47.75 feet
- B. -23.25 feet
- C. -22.75 feet
- D. 23.25 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q3/5

1348

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. -23.25 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q3/5

A hiker goes from an elevation of 1,200 feet to an elevation of -80 feet.

What is the change in elevation?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q4/5

1349

QUESTION ID

S36-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$-1,280$ feet

GRADING NOTE A decrease of 1,280 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q4/5

True or false?

If the temperature drops from -4°C to -9°C , it gets 5°C colder.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q5/5

1350

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 6.NS.C.7.c

FORWARD 2

Compare magnitudes with powers of 10 • Grade 8 • Expressions & Equations •
8.EE.A.3

One company's debt is 2×10^6 dollars. Another company's debt is 4×10^4 dollars.

How many times as large is the first debt?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q1/5

1352

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q1/5

The deepest ocean point is about -1×10^4 meters. A lake bottom is about -5×10^1 meters.

About how many times as deep is the ocean point?

- A.** 5 **B.** 20
C. 200 **D.** 2,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q2/5

1353

QUESTION ID

S36-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q2/5

Estimate 7,980,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q3/5

1354

QUESTION ID

S36-F2-Q3

TYPE

Short answer

CORRECT ANSWER

8×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q3/5

True or false?

6×10^9 is 300 times as large as 2×10^7 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q4/5

1355

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q4/5

Estimate 0.0000031 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q5/5

1356

QUESTION ID

S36-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3×10^{-6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q5/5

Distinguish comparisons of absolute value from statements about order

MAIN	Grade 6 main question	6.NS.C.7.d
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Ordering negative numbers	6.NS.C.7.a
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Compare the size and order of irrational numbers	8.NS.A.2

SET 37 • 6.NS.C.7.d

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.d

True or false?

An account balance less than $-\$50$ means a debt greater than $\$50$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q1/5

1359

QUESTION ID

S37-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q1/5

Which account balance shows the greatest debt?

- A.** −\$55
- B.** −\$38
- C.** −\$20
- D.** \$10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q2/5

1360

QUESTION ID

S37-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. −\$55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q2/5

True or false?

$-12 < -5$, so $|-12| < |-5|$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q3/5

1361

QUESTION ID

S37-M-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q3/5

Two temperatures are -15°F and -9°F .

Which temperature is colder? Which has the greater absolute value?

Colder: _____

Greater absolute value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q4/5

1362

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

**Colder: -15°F ;
Greater absolute
value: -15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q4/5

A submarine is at -300 feet. A diver is at -45 feet.

Which statement is true?

- A. The diver is deeper because $-45 > -300$.
- B. The submarine is deeper because $-300 > -45$.
- C. The submarine is deeper because $|-300| > |-45|$.
- D. They are at the same depth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q5/5

1363

QUESTION ID

S37-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The submarine is deeper because $|-300| > |-45|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q5/5

SET 37 • 6.NS.C.7.d

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-25|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q1/5

1365

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q1/5

Find the value.

|14|

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q2/5

1366

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q2/5

True or false?

$$|-3| = -3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q3/5

1367

QUESTION ID

S37-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q3/5

Which number has an absolute value of 7?

- A. -7
- B. $-\frac{1}{7}$
- C. 0
- D. $\frac{1}{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q4/5

1368

QUESTION ID

S37-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. -7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q4/5

Find the value.

$$|-2.5|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q5/5

1369

QUESTION ID

S37-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 6.NS.C.7.d

BACKWARD 2

Ordering negative numbers • Grade 6 • The Number System • 6.NS.C.7.a

Which number is least?

- A.** -20
- B.** -12
- C.** -5
- D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q1/5

1371

QUESTION ID

S37-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. -20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q1/5

True or false?

$$-9 > -4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q2/5

1372

QUESTION ID

S37-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Order from least to greatest.

$-3, -8, 1$

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q3/5

1373

QUESTION ID

S37-B2-Q3

TYPE

Short answer

CORRECT ANSWER

 $-8, -3, 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q3/5

1373

True or false?

$$-1 > -100$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q4/5

1374

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q4/5

Which number is greater?

A. -16

B. -6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q5/5

1375

QUESTION ID

S37-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. -6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 6.NS.C.7.d

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for a debt of \$40.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q1/5

1377

QUESTION ID

S37-B3-Q1

TYPE

Short answer

CORRECT ANSWER

−40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q1/5

What could -8 represent?

- A.** A deposit of \$8
- B.** A gain of 8 yards
- C.** 8 feet below sea level
- D.** 8 feet above sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q2/5

1378

QUESTION ID

S37-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 8 feet below sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q2/5

True or false?

An account balance of $-\$10$ means you owe $\$10$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q3/5

1379

QUESTION ID

S37-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q3/5

Write an integer for 15 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q4/5

1380

QUESTION ID

S37-B3-Q4

TYPE

Short answer

CORRECT ANSWER

-15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q4/5

True or false?

A 20-foot descent can be written as +20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q5/5

1381

QUESTION ID

S37-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • 6.NS.C.7.d

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

Two accounts have balances of $-\$42.50$ and $-\$18.75$.

How much greater is the larger debt?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q1/5

1383

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\$23.75$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q1/5

A diver at -24.5 m rises 9.75 m and then descends 15 m.

What is the diver's final position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q2/5

1384

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-29.75 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q2/5

The temperature at 6 a.m. was -8.5°F . At noon it was 3.5°F .

How much did the temperature rise?

- A.** -12°F **B.** -5°F
C. 5°F **D.** 12°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q3/5

1385

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q3/5

Maria's balance is $-\$36$. She pays back $\frac{1}{4}$ of what she owes.

What is her new balance?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q4/5

1386

QUESTION ID

S37-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$-\$27$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

A change from -12 m to -30 m is a change of -18 m.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q5/5

1387

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 6.NS.C.7.d

FORWARD 2

Compare the size and order of irrational numbers • Grade 8 • The Number System • 8.NS.A.2

Which number has the greater absolute value?

- A. 4.3
- B. $-\sqrt{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q1/5

1389

QUESTION ID

S37-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $-\sqrt{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q1/5

True or false?

$$-\sqrt{10} < -3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q2/5

1390

QUESTION ID

S37-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q2/5

Order from least to greatest.

$-\sqrt{5}$, -2.5 , -2

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q3/5

1391

QUESTION ID

S37-F2-Q3

TYPE

Short answer

CORRECT ANSWER

-2.5 , $-\sqrt{5}$, -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q3/5

Which number is farther from 0 on a number line?

A. 3.1

B. $-\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q4/5

1392

QUESTION ID

S37-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $-\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q4/5

Between which two consecutive integers is $-\sqrt{30}$?

Between: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q5/5

1393

QUESTION ID

S37-F2-Q5

TYPE

Short answer

CORRECT ANSWER

−6 and −5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q5/5

Find distances between points with the same first or second coordinate

MAIN	Grade 6 main question	6.NS.C.8
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Naming points on the coordinate plane	6.NS.C.6.c
BACKWARD 3	Distance between whole numbers	2.MD.B.6
FORWARD 1	Distance between rational numbers	7.NS.A.1.c
FORWARD 2	Distance between points using the Pythagorean Theorem	8.G.B.8

SET 38 • 6.NS.C.8

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-3, 4)$ and $(5, 4)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q1/5

1396

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q1/5

Find the distance between $(2, -6)$ and $(2, 5)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q2/5

1397

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q2/5

What is the distance between $(-7, -2)$ and $(-1, -2)$?

- A. -6
- B. 3
- C. 6
- D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q3/5

1398

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

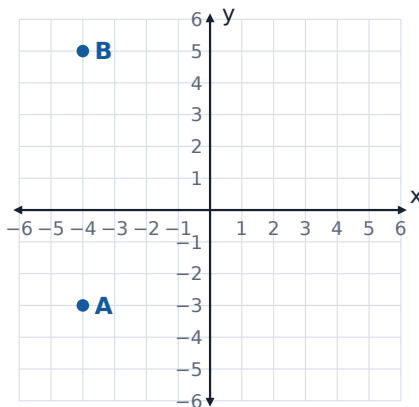
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q3/5

What is the distance between points A and B?

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q4/5

1399

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q4/5

Find the distance between $(-2.5, 1)$ and $(3.5, 1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q5/5

1400

QUESTION ID

S38-M-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q5/5

SET 38 • 6.NS.C.8

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-7|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q1/5

1402

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q1/5

True or false?

$$|-3| + |5| = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q2/5

1403

QUESTION ID

S38-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q2/5

$$|-4| + |6| = ?$$

- A.** -10
- B.** -2
- C.** 2
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q3/5

1404

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q3/5

Find the value.

$$|-9| + |2|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q4/5

1405

QUESTION ID

S38-B1-Q4

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q4/5

True or false?

The distance from -5 to 0 is $|-5|$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q5/5

1406

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q5/5

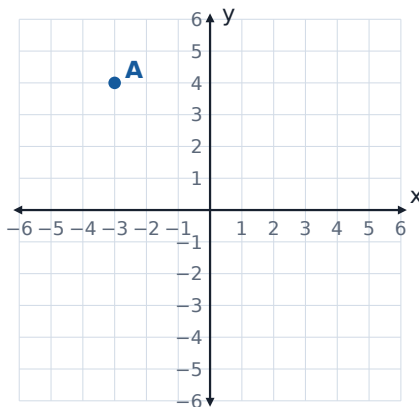
SET 38 • 6.NS.C.8

BACKWARD 2

Naming points on the coordinate plane • Grade 6 • The Number System •
6.NS.C.6.c

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q1/5

1408

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(-3, 4)

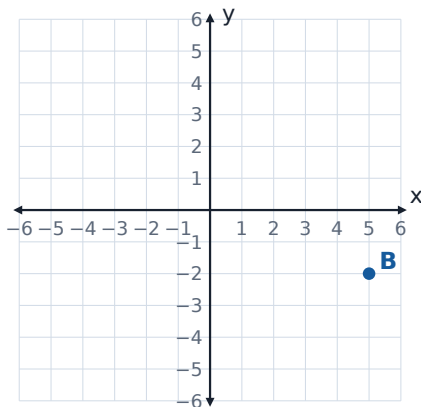
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q1/5

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q2/5

1409

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(5, -2)

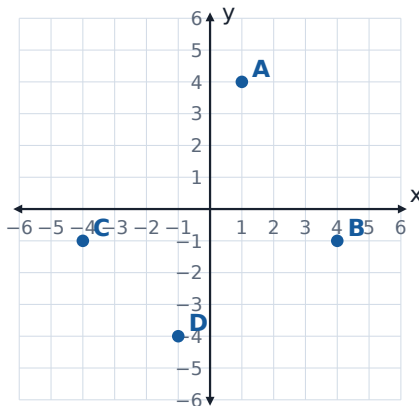
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q2/5

Which point is at $(-1, -4)$?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q3/5

1410

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

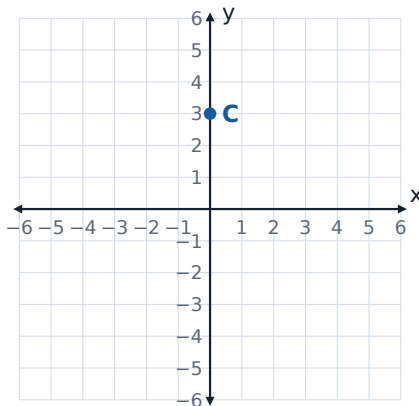
ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q3/5

True or false?

Point C is located at $(3, 0)$.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q4/5

1411

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

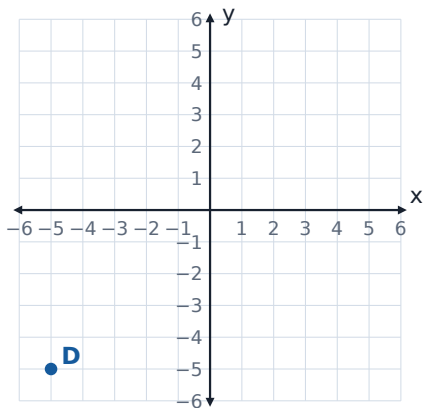
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q4/5

What are the coordinates of point D?

D = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q5/5

1412

QUESTION ID

S38-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(-5, -5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 6.NS.C.8

BACKWARD 3

Distance between whole numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 3 to 11 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q1/5

1414

QUESTION ID

S38-B3-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q1/5

How many units apart are points A and B?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q2/5

1415

QUESTION ID

S38-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q2/5

True or false?

The distance from 12 to 4 is 8 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q3/5

1416

QUESTION ID

S38-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q3/5

What is the distance from 5 to 14?

- A.** 8
- B.** 9
- C.** 10
- D.** 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q4/5

1417

QUESTION ID

S38-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q4/5

What is the distance from 0 to 13 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q5/5

1418

QUESTION ID

S38-B3-Q5

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • 6.NS.C.8

FORWARD 1

Distance between rational numbers • Grade 7 • The Number System •
7.NS.A.1.c

Find the distance between -2.5 and 4.5 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q1/5

1420

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q1/5

Find the value.

$$|-8.4 - 3.6|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q2/5

1421

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q2/5

What is the distance between $-3\frac{1}{2}$ and $2\frac{1}{4}$ on a number line?

A. $-5\frac{3}{4}$

B. $1\frac{1}{4}$

C. $5\frac{3}{4}$

D. $6\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q3/5

1422

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q3/5

True or false?

The distance between -7 and -2 is $|-7 - (-2)|$, which is 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q4/5

1423

QUESTION ID

S38-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q4/5

Find the distance between -4 and 2.5 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q5/5

1424

QUESTION ID

S38-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q5/5

SET 38 • 6.NS.C.8

FORWARD 2

Distance between points using the Pythagorean Theorem • Grade 8 •
Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q1/5

1426

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

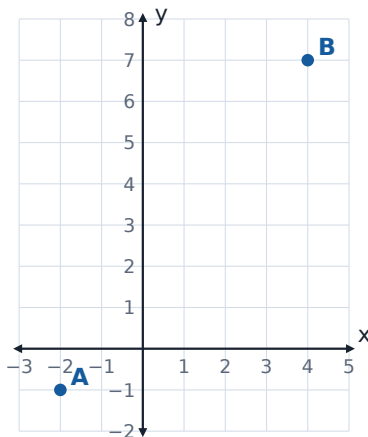
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q1/5

Find the distance between points A and B.

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q2/5

1427

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q2/5

What is the distance between (0, 0) and (5, 12)?

- A.** $\sqrt{119}$
- B.** 17
- C.** 13
- D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q3/5

1428

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q3/5

Find the distance between $(-3, 5)$ and $(1, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q4/5

1429

QUESTION ID

S38-F2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q4/5

True or false?

The distance between (2, 1) and (5, 5) is 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q5/5

1430

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q5/5

Solve real-world problems by graphing points in all four quadrants

MAIN	Grade 6 main question	6.NS.C.8
BACKWARD 1	Absolute value as distance	6.NS.C.7.c
BACKWARD 2	Plotting points in all four quadrants	6.NS.C.6.c
BACKWARD 3	Perimeter of rectangles	3.MD.D.8
BACKWARD 4	Area of rectangles	3.MD.C.7.b
FORWARD 1	Use a map scale on a coordinate grid	7.G.A.1
FORWARD 2	Straight-line distances on a map	8.G.B.8

SET 39 • 6.NS.C.8

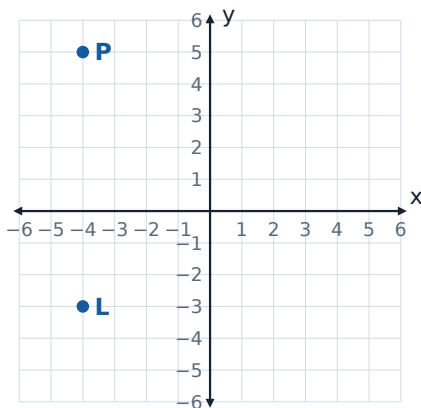
MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.8

On the map, each unit is 1 block.

How many blocks is it from the library L to the park P?

Blocks: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q1/5

1433

QUESTION ID

S39-M-Q1

TYPE

Short answer

CORRECT ANSWER

8 blocks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

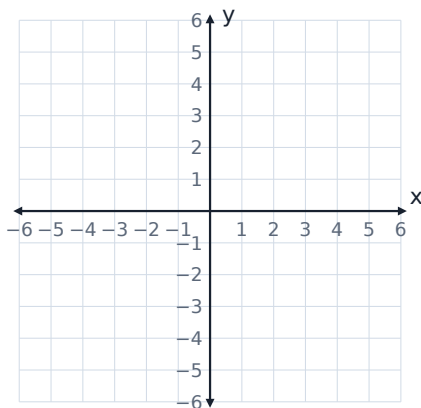
ID S39-M-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q1/5

A park is a rectangle with corners at $(-2, 3)$, $(4, 3)$, $(4, -1)$, and $(-2, -1)$. Each unit is 1 meter.

Graph the park. What is its perimeter?

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q2/5

1434

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

20 meters

GRADING NOTE Rectangle graphed with corners $(-2, 3)$, $(4, 3)$, $(4, -1)$, $(-2, -1)$; sides 6 m and 4 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

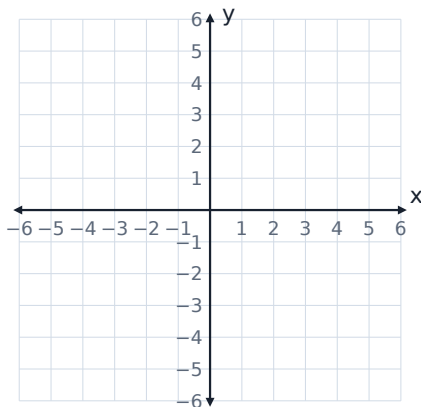
ID S39-M-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q2/5

The school is at $(-3, 2)$ and the pool is at $(-3, -4)$. Each unit is 1 block.

Graph both places. How many blocks apart are they?

Blocks: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q3/5

1435

QUESTION ID

S39-M-Q3

TYPE

Short answer

CORRECT ANSWER

6 blocks

GRADING NOTE Points graphed at $(-3, 2)$ and $(-3, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

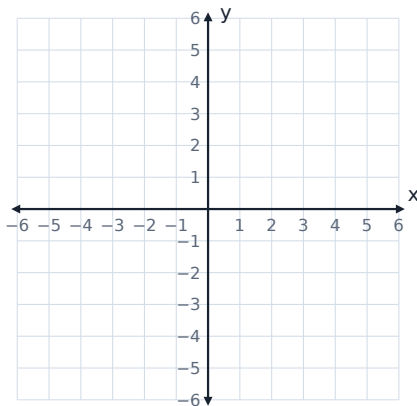
ID S39-M-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q3/5

A garden is a rectangle with corners at $(-5, 1)$, $(1, 1)$, $(1, -3)$, and $(-5, -3)$. Each unit is 1 meter.

What is the area of the garden?

- A.** 10 square meters **B.** 12 square meters
C. 20 square meters **D.** 24 square meters



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q4/5

1436

QUESTION ID

S39-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

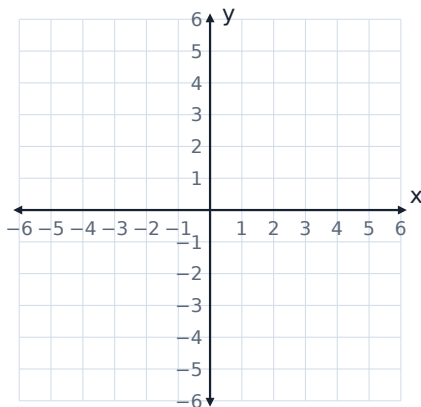
ID S39-M-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q4/5

A hiker starts at $(-5, 0)$, walks to $(3, 0)$, and then walks to $(3, 6)$. Each unit is 1 km.

Graph the path. How far does the hiker walk?

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q5/5

1437

QUESTION ID

S39-M-Q5

TYPE

Short answer

CORRECT ANSWER

14 km

GRADING NOTE Path graphed from $(-5, 0)$ to $(3, 0)$ to $(3, 6)$; $8 \text{ km} + 6 \text{ km}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q5/5

SET 39 • 6.NS.C.8

BACKWARD 1

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-4| + |5|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q1/5

1439

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q1/5

True or false?

The distance from -3 to 0 is 3 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q2/5

1440

QUESTION ID

S39-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q2/5

How far is -6 from 0?

- A.** -6
- B.** 0
- C.** 6
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q3/5

1441

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q3/5

Find the value.

$$|-2| + |4|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q4/5

1442

QUESTION ID

S39-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q4/5

True or false?

$$|-5| = -5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q5/5

1443

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

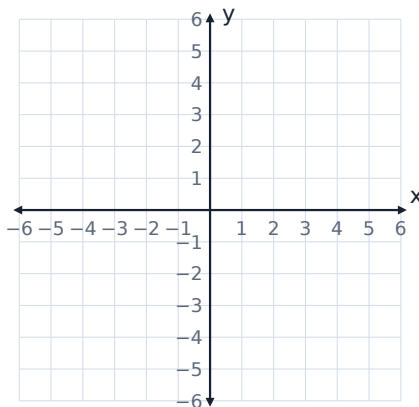
Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 6.NS.C.8

BACKWARD 2

Plotting points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

Plot and label point A($-2, 3$).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q1/5

1445

QUESTION ID

S39-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

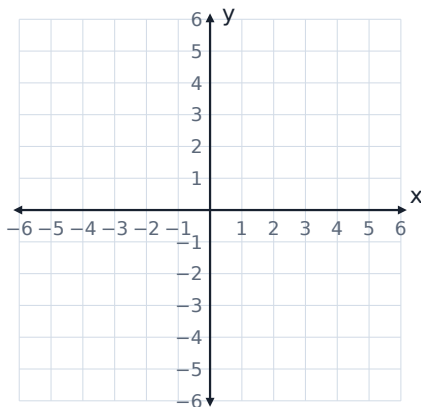
**Point A plotted at
($-2, 3$) and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q1/5

Plot and label point B(4, -1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q2/5

1446

QUESTION ID

S39-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Point B plotted at (4, -1) and labeled

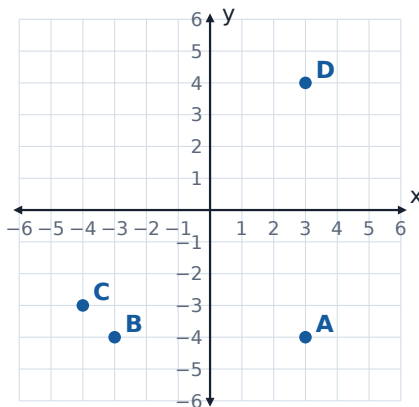
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q2/5

Which point is at $(-3, -4)$?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q3/5

1447

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

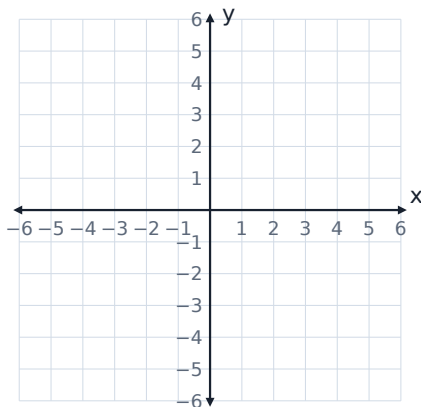
B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q3/5

Plot and label point C(0, -5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q4/5

1448

QUESTION ID

S39-B2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Point C plotted at (0, -5) on the y-axis and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

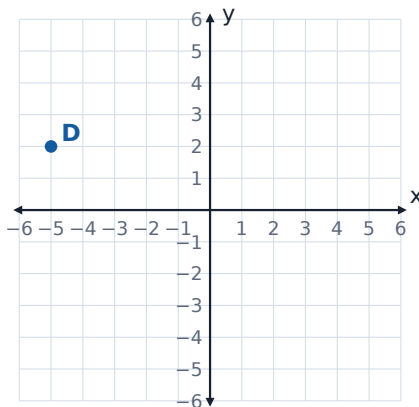
ID S39-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q4/5

True or false?

Point D is located at $(-5, 2)$.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q5/5

1449

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 6.NS.C.8

BACKWARD 3

Perimeter of rectangles • Grade 3 • Measurement & Data • 3.MD.D.8

What is the perimeter of the rectangle?

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q1/5

1451

QUESTION ID

S39-B3-Q1

TYPE

Short answer

CORRECT ANSWER

20 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q1/5

A rectangle is 9 cm long and 3 cm wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q2/5

1452

QUESTION ID

S39-B3-Q2

TYPE

Short answer

CORRECT ANSWER

24 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q2/5

True or false?

A square with 5-meter sides has a perimeter of 25 meters.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q3/5

1453

QUESTION ID

S39-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q3/5

A rectangle is 8 inches long and 2 inches wide.

What is its perimeter?

- A. 10 inches
- B. 16 inches
- C. 18 inches
- D. 20 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q4/5

1454

QUESTION ID

S39-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q4/5

A rectangle has a perimeter of 18 feet and a length of 5 feet.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q5/5

1455

QUESTION ID

S39-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q5/5

SET 39 • 6.NS.C.8

BACKWARD 4

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 6 m long and 4 m wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q1/5

1457

QUESTION ID

S39-B4-Q1

TYPE

Short answer

CORRECT ANSWER

24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q1/5

True or false?

A rectangle that is 7 units by 3 units has an area of 21 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q2/5

1458

QUESTION ID

S39-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q2/5

A rectangle is 5 ft by 8 ft.

What is its area?

- A.** 13 square feet
- B.** 26 square feet
- C.** 40 square feet
- D.** 80 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q3/5

1459

QUESTION ID

S39-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 40 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q3/5

What is the area of the rectangle?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q4/5

1460

QUESTION ID

S39-B4-Q4

TYPE

Short answer

CORRECT ANSWER

**18 square
centimeters**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q4/5

A square has 6-inch sides.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q5/5

1461

QUESTION ID

S39-B4-Q5

TYPE

Short answer

CORRECT ANSWER

36 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q5/5

SET 39 • 6.NS.C.8

FORWARD 1

Use a map scale on a coordinate grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is $\frac{1}{2}$ mile. The school is at $(-3, 2)$ and the park is at $(5, 2)$.

What is the actual distance between them?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q1/5

1463

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q1/5

On a trail map, each unit is 250 meters. The trail goes straight from $(0, -4)$ to $(0, 6)$.

How long is the actual trail?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q2/5

1464

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2,500 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q2/5

On a map, each unit is 1.5 km. A park has corners at $(-2, 1)$, $(4, 1)$, $(4, -3)$, and $(-2, -3)$.

What is the actual perimeter of the park?

- A.** 13.3 km **B.** 20 km
C. 30 km **D.** 36 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q3/5

1465

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 30 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q3/5

On a plan, each unit is 20 feet. A garden has corners at $(-1, -1)$, $(3, -1)$, $(3, 2)$, and $(-1, 2)$.

What are the actual length and width?

Length and width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q4/5

1466

QUESTION ID

S39-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**Length: 80 feet;
Width: 60 feet**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q4/5

True or false?

On a map where each unit is 5 miles, the points $(-3, 0)$ and $(3, 0)$ are 30 miles apart.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q5/5

1467

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 6.NS.C.8

FORWARD 2

Straight-line distances on a map • Grade 8 • Geometry • 8.G.B.8

On a map, the library is at $(-2, 1)$ and the pool is at $(4, 9)$. Each unit is 1 block.

What is the straight-line distance between them?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q1/5

1469

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10 blocks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q1/5

A drone flies from (0, 0) to (9, 12). Each unit is 1 meter.

How far does it fly in a straight line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q2/5

1470

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q2/5

Home is at $(-3, -1)$ and school is at $(1, 2)$. Each unit is 1 km.

What is the straight-line distance?

- A.** 5 km **B.** 25 km
C. 7 km **D.** $\sqrt{7}$ km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q3/5

1471

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q3/5

Find the straight-line distance from $(-4, 6)$ to $(2, -2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q4/5

1472

QUESTION ID

S39-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q4/5

True or false?

The straight-line distance from (1, 1) to (7, 9) is 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q5/5

1473

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q5/5

Evaluate numerical expressions with whole-number exponents

MAIN	Grade 6 main question	6.EE.A.1
BACKWARD 1	Powers of 10 with exponents	5.NBT.A.2
BACKWARD 2	Order of operations with parentheses	5.OA.A.1
BACKWARD 3	Repeated multiplication	3.OA.C.7
FORWARD 1	Products of signed numbers	7.NS.A.2.a
FORWARD 2	Evaluate expressions with integer exponents	8.EE.A.1

SET 40 • 6.EE.A.1

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2^4 + 3^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q1/5

1476

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q1/5

Evaluate.

$$3 \times 2^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q2/5

1477

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q2/5

Evaluate.

$$(5 - 2)^3 \div 9$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q3/5

1478

QUESTION ID

S40-M-Q3

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q3/5

Evaluate.

$$5^2 - 4 \times 2$$

- A. -3
- B. 17
- C. 21
- D. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q4/5

1479

QUESTION ID

S40-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q4/5

Evaluate.

$$\left(\frac{2}{3}\right)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q5/5

1480

QUESTION ID

S40-M-Q5

TYPE

Short answer

CORRECT ANSWER

4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q5/5

SET 40 • 6.EE.A.1

BACKWARD 1

Powers of 10 with exponents • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Write 10^3 as a whole number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q1/5

1482

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q1/5

True or false?

$$10^4 = 10,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q2/5

1483

QUESTION ID

S40-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q2/5

Which is equal to 100,000?

- A. 10^5
- B. 5^{10}
- C. 10^4
- D. 10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q3/5

1484

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q3/5

Write 1,000,000 as a power of 10.

Power of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q4/5

1485

QUESTION ID

S40-B1-Q4

TYPE

Short answer

CORRECT ANSWER

10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q4/5

True or false?

$$10^2 = 20$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q5/5

1486

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 6.EE.A.1

BACKWARD 2

Order of operations with parentheses • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$(8 + 4) \div 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q1/5

1488

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q1/5

Evaluate.

$$20 - (3 \times 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q2/5

1489

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q2/5

True or false?

$$2 \times (3 + 5) = 11$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q3/5

1490

QUESTION ID

S40-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q3/5

Evaluate.

$$18 \div (2 + 4)$$

- A.** 3
- B.** 6
- C.** 9
- D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q4/5

1491

QUESTION ID

S40-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q4/5

Evaluate.

$$(9 - 3) \times 5$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q5/5

1492

QUESTION ID

S40-B2-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 6.EE.A.1

BACKWARD 3

Repeated multiplication • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$2 \times 2 \times 2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q1/5

1494

QUESTION ID

S40-B3-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q1/5

1494

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$5 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q2/5

1495

QUESTION ID

S40-B3-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q2/5

1495

True or false?

$$3 \times 3 \times 3 = 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q3/5

1496

QUESTION ID

S40-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q3/5

$$4 \times 4 \times 4 = ?$$

- A.** 12
- B.** 16
- C.** 44
- D.** 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q4/5

1497

QUESTION ID

S40-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$6 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q5/5

1498

QUESTION ID

S40-B3-Q5

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q5/5

1498

SET 40 • 6.EE.A.1

FORWARD 1

Products of signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Evaluate.

$$(-3)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q1/5

1500

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Evaluate.

$$(-2)^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q2/5

1501

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q2/5

1501

True or false?

$$-3^2 = (-3)^2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q3/5

1502

QUESTION ID

S40-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q3/5

$$(-1)^5 = ?$$

- A.** -5
- B.** -1
- C.** 1
- D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q4/5

1503

QUESTION ID

S40-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q4/5

Evaluate.

$$\left(-\frac{1}{2}\right)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q5/5

1504

QUESTION ID

S40-F1-Q5

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q5/5

SET 40 • 6.EE.A.1

FORWARD 2

Evaluate expressions with integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Evaluate.

$$2^{-3}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q1/5

1506

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q1/5

Evaluate.

$$5^{-2} \times 5^4$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q2/5

1507

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q2/5

$$10^{-2} = ?$$

- A. -100
- B. -20
- C. $\frac{1}{100}$
- D. 0.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q3/5

1508

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q3/5

Evaluate.

$$\frac{4^5}{4^3}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q4/5

1509

QUESTION ID

S40-F2-Q4

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q4/5

True or false?

$$7^0 = 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q5/5

1510

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q5/5

Write numerical expressions with whole-number exponents

MAIN	Grade 6 main question	6.EE.A.1
BACKWARD 1	Powers of 10 with exponents	5.NBT.A.2
BACKWARD 2	Multiplying with repeated factors	3.OA.C.7
FORWARD 1	Powers of negative numbers	7.NS.A.2.a
FORWARD 2	Write expressions as a single power	8.EE.A.1

SET 41 • 6.EE.A.1

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.1

Write using an exponent.

$$7 \times 7 \times 7 \times 7$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q1/5

1513

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

7⁴

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q1/5

Which expression is equal to $3 \times 3 \times 3 \times 3 \times 3$?

- A. 5^3
- B. 15
- C. 3^5
- D. 3×5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q2/5

1514

QUESTION ID

S41-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 3^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q2/5

Write using an exponent.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q3/5

1515

QUESTION ID

S41-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(1/2)^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q3/5

Write using an exponent.

$$0.4 \times 0.4$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q4/5

1516

QUESTION ID

S41-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.4^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q4/5

Write using exponents.

$$2 \times 2 \times 2 \times 5 \times 5$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q5/5

1517

QUESTION ID

S41-M-Q5

TYPE

Short answer

CORRECT ANSWER

$$2^3 \times 5^2$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q5/5

SET 41 • 6.EE.A.1

BACKWARD 1

Powers of 10 with exponents • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Write $10 \times 10 \times 10$ as a power of 10.

Power of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q1/5

1519

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q1/5

True or false?

$$10 \times 10 \times 10 \times 10 = 10^4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q2/5

1520

QUESTION ID

S41-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q2/5

Which is equal to 10^5 ?

- A. $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$
- B. $10 \times 10 \times 10 \times 10 \times 10$
- C. 100,000,000
- D. 10×5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q3/5

1521

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B.

$10 \times 10 \times 10 \times 10 \times 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q3/5

How many factors of 10 are in 10^6 ?

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q4/5

1522

QUESTION ID

S41-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q4/5

True or false?

$$10^2 = 10 \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q5/5

1523

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 6.EE.A.1

BACKWARD 2

Multiplying with repeated factors • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$2 \times 2 \times 2 \times 2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q1/5

1525

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q1/5

1525

Multiply.

$$3 \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q2/5

1526

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q2/5

True or false?

$$5 \times 5 \times 5 = 15$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q3/5

1527

QUESTION ID

S41-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q3/5

How many times is 7 used as a factor in $7 \times 7 \times 7$?

- A. 1
- B. 3
- C. 7
- D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q4/5

1528

QUESTION ID

S41-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4 \times 4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q5/5

1529

QUESTION ID

S41-B2-Q5

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q5/5

1529

SET 41 • 6.EE.A.1

FORWARD 1

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2.a

Write using an exponent.

$$(-3)(-3)(-3)(-3)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q1/5

1531

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(-3)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q1/5

Write $(-2)(-2)(-2)$ using an exponent. Is its value positive or negative?

Expression: _____

Sign: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q2/5

1532

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(-2)^3$; negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q2/5

Which expression is equal to $(-5)^2$?

- A. -5×2
- B. $(-5)(-5)$
- C. -5×5
- D. $-(5 \times 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q3/5

1533

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(-5)(-5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q3/5

True or false?

$(-1)^6$ is positive.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q4/5

1534

QUESTION ID

S41-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q4/5

Write using an exponent.

$$\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q5/5

1535

QUESTION ID

S41-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$\left(-\frac{2}{3}\right)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q5/5

SET 41 • 6.EE.A.1

FORWARD 2

Write expressions as a single power • Grade 8 • Expressions & Equations •
8.EE.A.1

Write as a single power.

$$4^3 \times 4^5$$

Single power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q1/5

1537

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4⁸

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q1/5

Write as a single power.

$$(2^3)^4$$

Single power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q2/5

1538

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2¹²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q2/5

Which is equal to $\frac{6^7}{6^2}$?

- A. 6^{14}
- B. 6^5
- C. 6^9
- D. 1^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q3/5

1539

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q3/5

Write as a single power.

$$3^4 \times 3^{-6}$$

Single power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q4/5

1540

QUESTION ID

S41-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3^{-2}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q4/5

True or false?

$$5^2 \times 5^3 = 5^6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q5/5

1541

QUESTION ID

S41-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q5/5

Write expressions in which letters stand for numbers

MAIN	Grade 6 main question	6.EE.A.2.a
BACKWARD 1	Words for operations	5.OA.A.2
BACKWARD 2	A symbol for an unknown number	3.OA.A.4
BACKWARD 3	Order matters in subtraction and division	3.OA.B.5
FORWARD 1	Write and solve equations of the form $px + q = r$	7.EE.B.4.a
FORWARD 2	Construct a function from a description	8.F.B.4

SET 42 • 6.EE.A.2.a

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.a

Write an expression for "subtract m from 12."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q1/5

1544

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

12 – m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q1/5

Write an expression for "5 more than twice a number n ."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q2/5

1545

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

$2n + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q2/5

Which expression means "the quotient of x and 4"?

A. $4x$

B. $\frac{4}{x}$

C. $x - 4$

D. $\frac{x}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q3/5

1546

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q3/5

Write an expression for "3 less than the product of 7 and a number m ."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q4/5

1547

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

$7m - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q4/5

True or false?

"8 less than k" can be written as $8 - k$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q5/5

1548

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q5/5

SET 42 • 6.EE.A.2.a

BACKWARD 1

Words for operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Which expression shows "add 6 and 9, then multiply by 4"?

- A. $6 + 9 \times 4$
- B. $4 \times (6 + 9)$
- C. $(4 + 6) \times 9$
- D. $4 \times 6 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q1/5

1550

QUESTION ID

S42-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $4 \times (6 + 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q1/5

Write an expression for "subtract 4 from 10, then divide by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q2/5

1551

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(10 - 4) \div 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q2/5

True or false?

"The product of 6 and 9" means 6×9 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q3/5

1552

QUESTION ID

S42-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q3/5

Which word describes the answer to a subtraction problem?

- A.** Quotient
- B.** Difference
- C.** Product
- D.** Sum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q4/5

1553

QUESTION ID

S42-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Difference

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q4/5

Write an expression for "the sum of 12 and 5, multiplied by 3."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q5/5

1554

QUESTION ID

S42-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(12 + 5) \times 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 6.EE.A.2.a

BACKWARD 2

A symbol for an unknown number • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.4

What number goes in the box?

$$\square + 7 = 12$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q1/5

1556

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q1/5

True or false?

In $7 \times \square = 42$, the box stands for 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q2/5

1557

QUESTION ID

S42-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q2/5

$$\square - 5 = 9$$

What number goes in the box?

- A. 4
- B. 9
- C. 14
- D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q3/5

1558

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q3/5

What number goes in the box?

$$20 \div \square = 4$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q4/5

1559

QUESTION ID

S42-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q4/5

What number goes in the box?

$$\square \times 3 = 27$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q5/5

1560

QUESTION ID

S42-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 6.EE.A.2.a

BACKWARD 3

Order matters in subtraction and division • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$7 - 3 = 3 - 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q1/5

1562

QUESTION ID

S42-B3-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q1/5

True or false?

$$6 \times 4 = 4 \times 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q2/5

1563

QUESTION ID

S42-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q2/5

True or false?

$$12 \div 3 = 3 \div 12$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q3/5

1564

QUESTION ID

S42-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q3/5

Which statement is true?

- A. $10 \div 2 = 2 \div 10$
- B. $5 + 9 = 9 + 5$
- C. $8 - 2 = 2 - 8$
- D. $9 - 5 = 5 - 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q4/5

1565

QUESTION ID

S42-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $5 + 9 = 9 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q4/5

True or false?

$$3 + 11 = 11 + 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q5/5

1566

QUESTION ID

S42-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • 6.EE.A.2.a

FORWARD 1

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A rectangle has a perimeter of 46 inches. Its width is 8 inches.

Write and solve an equation to find its length.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q1/5

1568

QUESTION ID

S42-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**$2l + 2(8) = 46; l = 15$
inches**

GRADING NOTE Any equivalent equation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q1/5

Jo has \$40. She buys 3 shirts that cost the same amount and has \$4 left.

How much does each shirt cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q2/5

1569

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

\$12

GRADING NOTE $3s + 4 = 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q2/5

Which equation represents "5 more than 3 times a number is 26"?

- A. $3(n + 5) = 26$
- B. $5n + 3 = 26$
- C. $3n + 5 = 26$
- D. $3n - 5 = 26$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q3/5

1570

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $3n + 5 = 26$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q3/5

Twice a number minus 7 is 15.

Write and solve an equation to find the number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q4/5

1571

QUESTION ID

S42-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$2n - 7 = 15; n = 11$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q4/5

A phone plan costs \$30 per month plus \$0.10 per text.
One month's bill is \$42.

How many texts were sent?

Texts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q5/5

1572

QUESTION ID

S42-F1-Q5

TYPE

Short answer

CORRECT ANSWER

120 texts

GRADING NOTE $30 + 0.10t = 42$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q5/5

SET 42 • 6.EE.A.2.a

FORWARD 2

Construct a function from a description • Grade 8 • Functions • 8.F.B.4

A pool has 500 gallons of water. It is filled at 25 gallons per minute.

Write a function for the gallons G after t minutes.

$G =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q1/5

1574

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$G = 500 + 25t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q1/5

Tickets cost \$4 each plus a \$6 fee for the order.

Which function gives the total cost y for x tickets?

- A. $y = 10x$
- B. $y = 4x + 6$
- C. $y = 6x + 4$
- D. $y = 4x - 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q2/5

1575

QUESTION ID

S42-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 4x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q2/5

The table shows a linear function.

Write an equation for the function.

x	0	1	2
y	7	10	13

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q3/5

1576

QUESTION ID

S42-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$$y = 3x + 7$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q3/5

A line passes through $(0, -2)$ and $(3, 4)$.

Write an equation for the line.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q4/5

1577

QUESTION ID

S42-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$y = 2x - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q4/5

True or false?

The function $y = 15x + 20$ has an initial value of 15.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q5/5

1578

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q5/5

Identify parts of an expression: term, sum, product, factor, quotient, coefficient

MAIN	Grade 6 main question	6.EE.A.2.b
BACKWARD 1	Sum, difference, product, and quotient	5.OA.A.2
BACKWARD 2	Factors	4.OA.B.4
BACKWARD 3	Interpreting expressions without evaluating	5.OA.A.2
FORWARD 1	Combine like terms with rational coefficients	7.EE.A.1
FORWARD 2	Interpret the parts of a linear function	8.F.B.4

SET 43 • 6.EE.A.2.b

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.b

In the expression $4(9 + 3)$, how can the 4 be described?

- A.** A sum
- B.** A factor
- C.** A term of $9 + 3$
- D.** A quotient

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q1/5

1581

QUESTION ID

S43-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. A factor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q1/5

How many terms are in the expression $3x + 5y - 2$?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q2/5

1582

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q2/5

What is the coefficient of y in $4x + 9y$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q3/5

1583

QUESTION ID

S43-M-Q3

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q3/5

True or false?

In $5(m + n)$, the quantity $(m + n)$ is a single factor and also a sum of two terms.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q4/5

1584

QUESTION ID

S43-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q4/5

In the expression $7m + 3$, which part is the constant term?

- A. 3
- B. $7m$
- C. m
- D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q5/5

1585

QUESTION ID

S43-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q5/5

SET 43 • 6.EE.A.2.b

BACKWARD 1

Sum, difference, product, and quotient • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

8 + 7 is a sum.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q1/5

1587

QUESTION ID

S43-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q1/5

What is 6×9 called?

- A.** A quotient
- B.** A product
- C.** A sum
- D.** A difference

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q2/5

1588

QUESTION ID

S43-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. A product

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q2/5

Is $24 \div 6$ a sum, difference, product, or quotient?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q3/5

1589

QUESTION ID

S43-B1-Q3

TYPE

Short answer

CORRECT ANSWER

Quotient

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q3/5

True or false?

15 – 4 is called a product.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q4/5

1590

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q4/5

Which expression is a quotient?

- A. 20×5
- B. $20 + 5$
- C. $20 - 5$
- D. $20 \div 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q5/5

1591

QUESTION ID

S43-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $20 \div 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 6.EE.A.2.b

BACKWARD 2

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

In $4 \times 7 = 28$, the numbers 4 and 7 are factors of 28.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q1/5

1593

QUESTION ID

S43-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q1/5

Name the two factors in 5×9 .

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q2/5

1594

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5 and 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q2/5

Which number is NOT a factor of 20?

- A.** 4
- B.** 5
- C.** 6
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q3/5

1595

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q3/5

True or false?

3 is a factor of 12.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q4/5

1596

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q4/5

Write one factor pair of 18.

Factor pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q5/5

1597

QUESTION ID

S43-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1 and 18

GRADING NOTE Also correct: 2 and 9, or 3 and 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 6.EE.A.2.b

BACKWARD 3

Interpreting expressions without evaluating • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$4 \times (2,365 + 718)$ is four times as large as $2,365 + 718$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q1/5

1599

QUESTION ID

S43-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q1/5

How does $5 \times (26 + 4)$ compare to $26 + 4$?

- A. It is the same.
- B. It is 5 times as large.
- C. It is 5 more.
- D. It is 5 less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q2/5

1600

QUESTION ID

S43-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. It is 5 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q2/5